



OCP Accelerator Module Design Specification v1.0

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1 Contents

2	License.....	5
3	Acknowledge.....	6
4	Overview	7
4.1	Scope.....	8
4.2	Acronym.....	8
5	High Level Specification for the OCP Acceleration Module	9
6	Accelerator Module Mechanical Specifications.....	10
6.1	Module PCBA Form Factor.....	10
6.2	Mezzanine Connector	13
6.2.1	Mate/Unmate Force Data	14
6.3	Accelerator Module Bottom Stiffener	15
6.3.1	Tolerance Stack-up of Bottom Stiffener	18
6.3.2	Alignment pins	18
6.3.3	EMI Gaskets / Pads.....	19
6.3.4	Die Spring	20
6.4	Baseboard Keepout Zone & Grounding Pads.....	22
6.4.1	SMT Nut	23
6.4.2	Component Keep-out Zone.....	23
6.4.3	Grounding Pads.....	23
6.5	Recommended Alignment Features	23
6.6	Reference Heatsink Design	25
6.6.1	Top Handle	25
6.6.2	Long Screw Attachment	26
7	Thermal Specification	27
7.1	Environmental Conditions	27
7.2	Temperature Report	28
7.2.1	Temperature Sensors.....	28
7.2.2	Remaining Components.....	28
7.3	Thermal module info.....	28
7.4	Heatsink Assembly	28

7.5	Thermal Recommendation	29
7.5.1	Airflow Budget	29
7.5.2	Reference Heatsink Design	29
7.5.3	Cooling Limit	31
7.5.4	Heatsink Installation	33
7.5.5	Thermal Interface Material	34
8	Accelerator Module Electrical Specification	36
8.1	Electrical Connector	36
8.2	Mezzanine Module Connector Pinout Quadrants	37
8.3	Accelerator Module Pinout Description	38
8.4	Accelerator Module Power Profiles	47
8.4.1	Thermal Design Power TDP	47
8.4.2	Excursion Design Power EDP	48
8.5	System power sequencing	48
8.6	Mezzanine Module Insertion Loss	50
9	Mezzanine Module Interconnect Topologies	52
9.1	Module ID	52
9.2	Interconnect Topology	53
9.2.1	Hybrid Cube Mesh (HCM)	53
9.2.2	Almost Fully Connected	54
9.2.3	Fully Connected	55
9.2.4	4D Hypercube	57
9.3	LINK_CONFIG[4:0]	58
9.4	Accelerator Interconnect PCB Topology	59
10	Mezzanine module reference system design	60
11	System Level Communication	61
11.1	Sideband Management Interface	61
11.2	Sensor Reporting	61
11.3	Error Monitoring/Reporting	62
11.4	Firmware Update	62
11.5	Power Capping	62

11.6	FRU Information.....	62
11.7	IO Calibration	62
12	Environmental.....	64
12.1	Environmental Requirements	64
12.2	Regulation	64
13	Revision History	65

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3 Acknowledge

We would like to acknowledge **Intel Corporation** and Advanced Micro Devices(**AMD**), Inc. for their collaboration on the engineering analysis for the OCP accelerator module specification development, including physical form factor, logical connectivity, signal breakout and routing studies, signal integrity analysis, and power delivery analysis.

A special acknowledgement to Molex LLC for their cross-functional support as it pertains to the Mirror Mezz connector and its implementation into this module specification.

We also want to acknowledge the community's great support for specification. After we first time talked the common form factor accelerator module during the OCP server group monthly call and HPC group monthly call in November 2018, we got a lot great feedback from the community, for us to enhance the specification can be adopted by wider community.

4 Overview

Artificial Intelligence (AI) applications are rapidly evolving and producing an explosion of new types of hardware accelerators for Machine Learning (ML), Deep Learning (DL), and High-performance Computing (HPC).

Different implementations target similar requirements for power/cooling, robustness, serviceability, configuration, programming, management, debug, inter-module communication to scale-up, and input/output bandwidth to scale-out.

To take advantage of the available industry-standard form factors to reduce the required time and effort in producing suitable solutions, various implementations have selected PCIe CEM form factor as a quick market entry.

Such solutions are not optimized for the upcoming AI workloads which require ever-growing bandwidth and interconnect flexibility for data/model parallelism.

The state-of-the-art applications require multiple cards in a system with multiple inter-card links running at highspeed interconnect bandwidth between cards.

Using PCIe CEM form factor to meet such interconnect requirement poses several challenges such as excessive signal insertion loss from ASIC to PCIe connectors and on baseboard; inter-card cabling complexity reducing robustness and serviceability; and limits the supported inter-ASIC topologies.

To enable flexible high-speed interconnect topologies for multi-ASIC solutions, this base specification outlines an interoperable, modular hierarchy based on a mezzanine module form factor (OCP Accelerator Module: OAM), an interconnect Baseboard, a Tray, and a Chassis.

- Mezzanine Module (various accelerators)
- Baseboard (interconnecting topologies between accelerators, hosts, and other IO devices to scale up)
- Tray (a means for ease of field replacement and serviceability)
- Chassis (an outline for a collection of Trays and input/output resources to scale out)

Based on this base specification, various design and product implementations may maintain interoperability while offering enhancements in each hierarchy level.

We invite open contributions in the following areas:

1. Base specification (OCP Accelerator Infrastructure Project Specification)
2. Design specification (This document, detailed description of alternative, interoperable components which meet the base specification)
3. Products (schematic, layout, mechanical/thermal solutions, and firmware/software to realize the above designs)

4.1 Scope

The OCP Accelerator Mezzanine Module design specification defines the form factor and common specifications for a compute accelerator module and a compliant base board design enabling interoperability across multiple ASIC or GPU based mezzanine modules and a based board design interface.

The mezzanine module form factor facilitates scalability across accelerators by simplifying the system solution when interconnecting communication links among modules in comparison with a PCIe Add-in card form factor.

4.2 Acronyms

Acronym	Definition
ASIC	Application Specific Integrated Circuit
OAM	OCP Accelerator Module
BGA	Ball Grid Array
BMC	Baseboard Management Controller
TDP	Thermal Design Power
EDP	Excursion Design Power
GPU	Graphic Processing Unit
MPN	Manufacturing Part Number
DXF	Drawing eXchange Format
PCBA	Printed Circuit Board Assembly

5 High Level Specification for the OCP Acceleration Module

Module Dimension	102mm x 165mm
Board Thickness	1.57 - 3.20mm $\pm$ 10%
Module Power/Input Voltage	○ High power module supports up to 700W, using 44V-59.5V DC as input power ○ Low power module supports up to 350W, using 11-13.2V DC as input power
Connectors	2* Molex Mirror Mezz Connectors (MPN: 209311-1115) Stack height 5mm Differential pair Impedance: 90ohm $\pm$ 5%
Host Interface	One or two x16 host link. E.g. PCIe Gen3/4/5 x16, or alternate protocols.
Module to Module Interconnect Links	Up to 7 Links per module, each link has up to X16-X20 lanes Each link may be able to be configured to sub links.
Bottom stiffener height (including Mylar)	5 $\pm$ 0.15mm

6 Accelerator Module Mechanical Specifications

The accelerator module form factor is described in this section. It uses a single accelerator ASIC on the module as an example to describe the mechanical specifications. The top and bottom stiffeners may be different if the modules have multiple accelerator ASICs.

Please refer to 2D DXF and 3D files for further details. 2D DXF and 3D files are included with the contribution package as well as relevant reference drawings to mechanical components. Please note that some features on the OAM are called out as required, but others are included merely for reference.

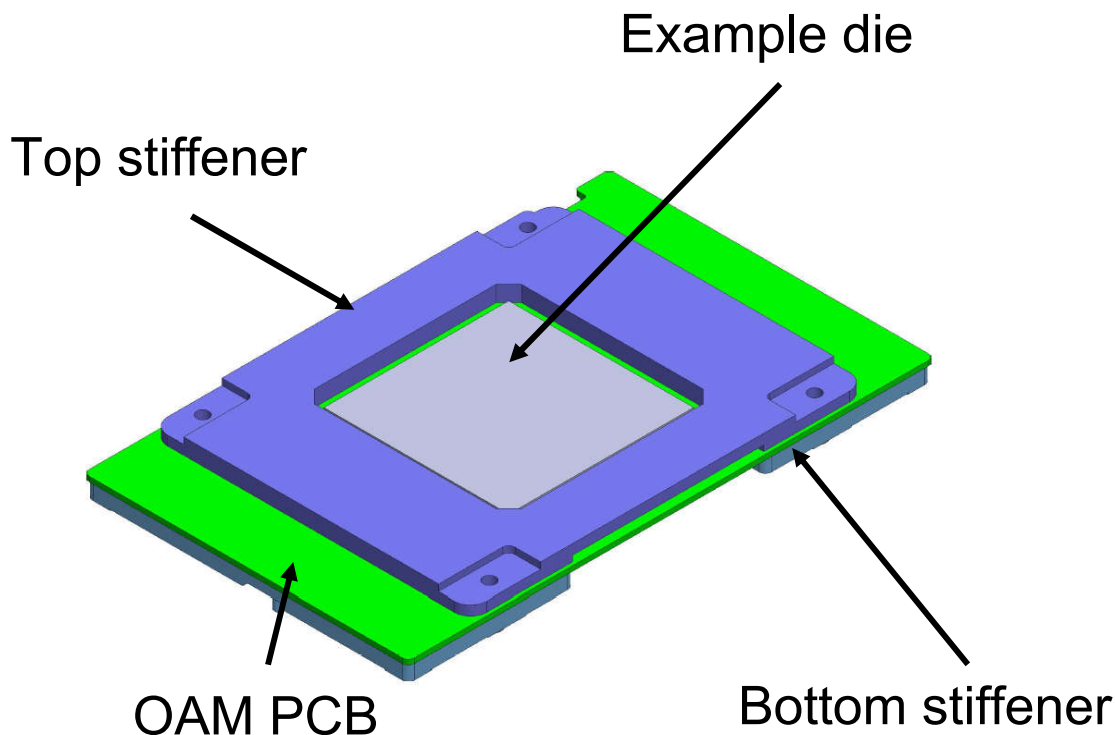


Figure 1 OCP Accelerator Module isometric view

6.1 Module PCBA Form Factor

This section covers the required and recommended dimensions of the module PCBA and its individual parts. Figures 2 and 3 illustrate Mezzanine form factor and dimensions, with all the dimensions in Figure 2 required. It is a 102mm x 165mm PCB size with Mezzanine Connectors on the bottom side and Accelerator on top side. Connector to connector pitch is 102mm. Four NPTH mounting holes are used to attach the module to a corresponding bolster plate secured below the system PCB. These mounting holes should provide clearance for a M3.5 screw, with enough thread length to secure to the bottom stiffener. There is a notch located near the southwest corner of the board, adjacent to Connector 1. For connector orientation, see Figure 4 Top and bottom views of the Accelerator Module Assembly.

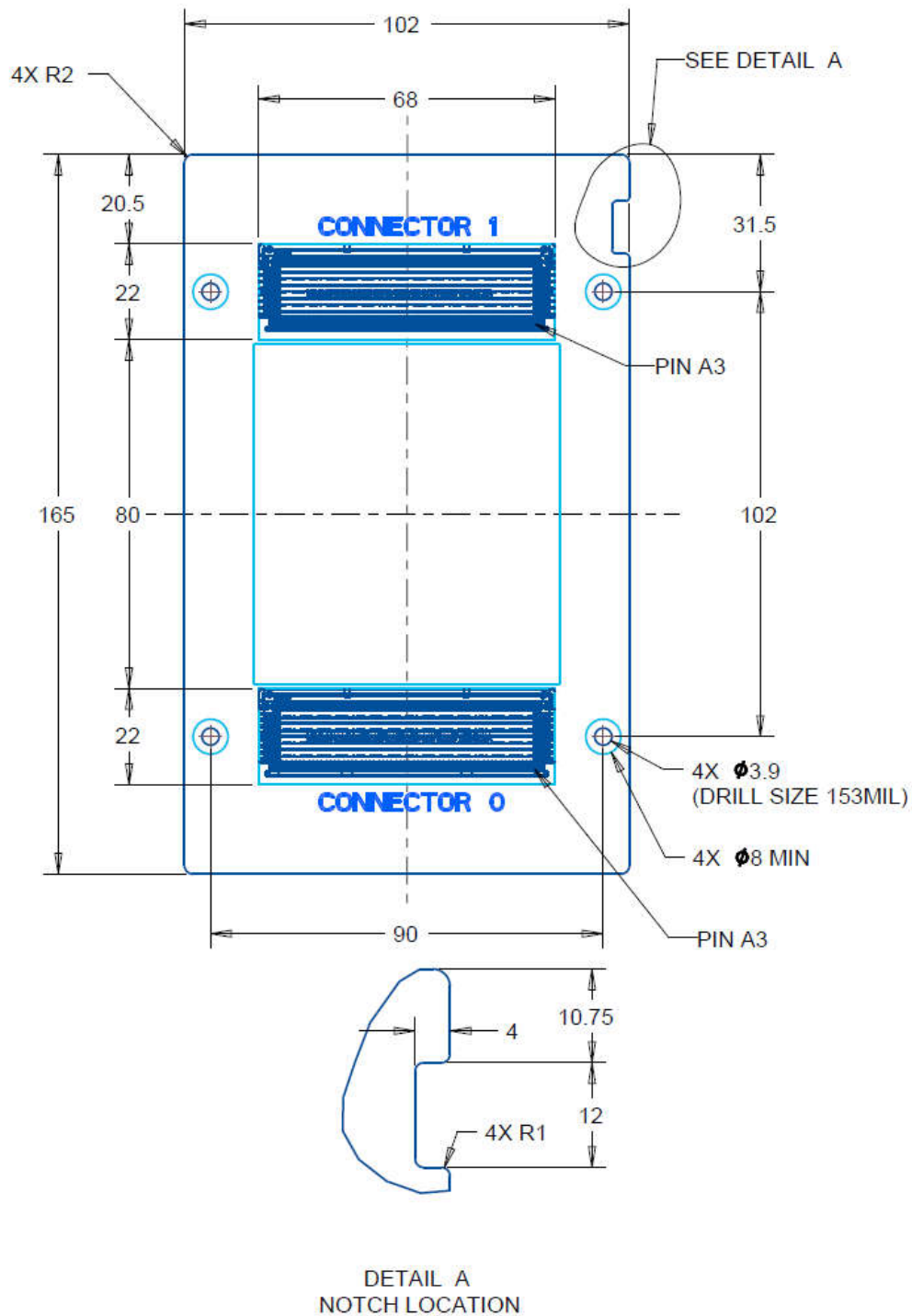


Figure 2 102mm wide OCP Accelerator Module Form Factor Dimensions, Bottom View



Figure 3 OCP Accelerator Module Form Factor, Side View with System Baseboard

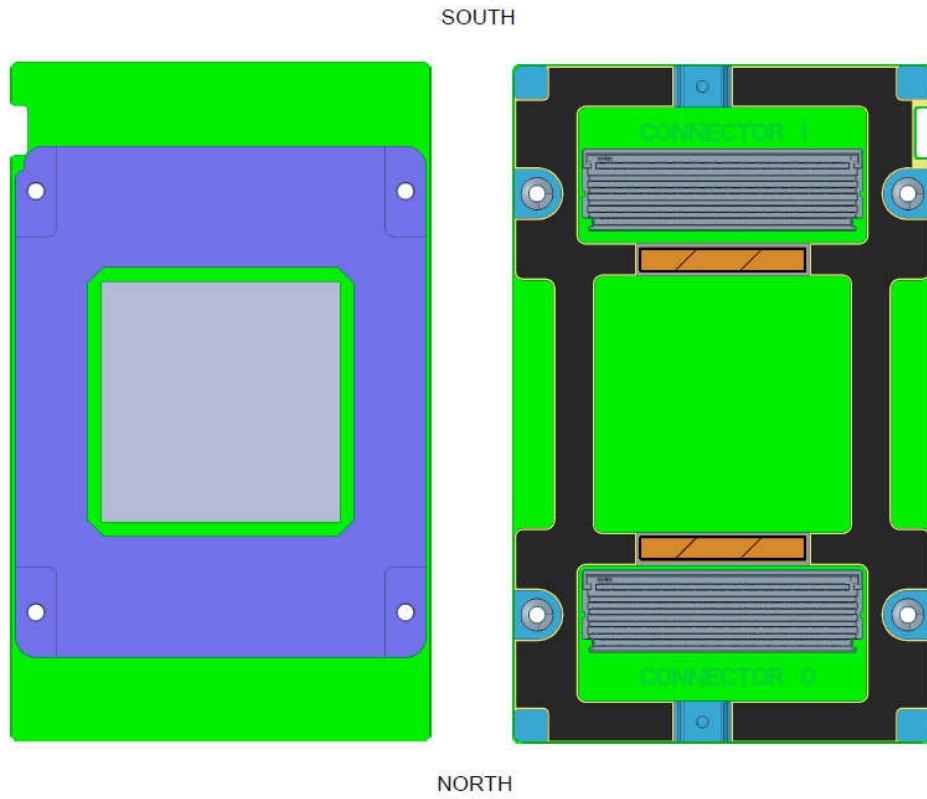


Figure 4 Top and bottom views of the Accelerator Module Assembly

6.2 Mezzanine Connector

Molex Mirror Mezz (MPN: 209311-1115) is the PCB to PCB interconnect solution supported by the OCP Accelerator Module form factor. Mirror Mezz is a highspeed differential pair-based mezzanine connector in a footprint-identical genderless plug and receptacle part for module and base board. Figure 5 Mirror Mezz 209311-1115 is provided courtesy of Molex.

- Stack Height: 5mm
- Mating Force: 0.35N/pin Max, total 240.8N MAX. Data on mate forces of the 209311-1115 connector is shown in Table 1 Mate/Unmate Averaged Data for Molex Mirror Mezz 209311-1115.
- Unmating force: 0.045N/pin MIN, total 31.0N MIN. Data on unmate forces of the 209311-1115 connector is shown in Table 1 Mate/Unmate Averaged Data for Molex Mirror Mezz 209311-1115.
- Weight of OAM + Heatsink: 2kg MAX
- 172 Total Differential Pairs, of which 161 are fully ground shielded (non-orphan)



Figure 5 Mirror Mezz 209311-1115

6.2.1 Mate/Unmate Force Data

The mating connectors will be vertically inverted when mated.

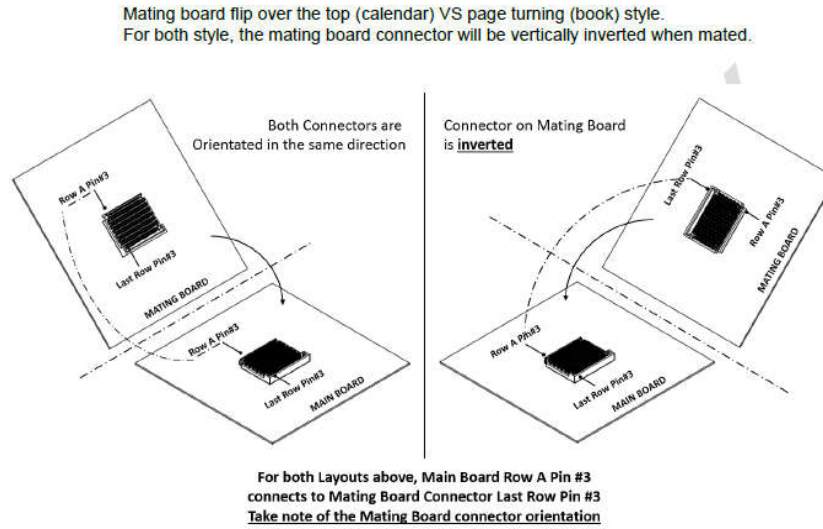


Figure 6 Mirror Mezz Connector Mating

The mate and unmate forces provided in the product specification are conservative. The specific 209311-1115 connector that the OAM uses has mate/unmate forces more in line with those found in Table 1 Mate/Unmate Averaged Data for Molex Mirror Mezz 209311-1115 and in Figure 7 Measured Mate Force per Pin for Molex Mirror Mezz 209311-1115. Note that the mate force per pin trends upwards for initial 5 cycles before settling back towards the average of 0.21N/pin.

Table 1 Mate/Unmate Averaged Data for Molex Mirror Mezz 209311-1115

Unit: N	Cycle	1st	2nd	3rd	4th	5th	Max	Min	Ave
Mating Force	Sample 1	161.5	168.9	177.9	178.0	178.7	182.8	153.2	167.6
	Sample 2	155.9	162.0	169.7	171.8	179.6			
	Sample 3	156.1	156.7	166.3	167.7	177.9			
	Sample 4	156.1	165.3	171.1	175.9	179.4			
	Sample 5	160.4	153.8	168.8	170.4	160.3			
	Sample 6	154.7	163.5	163.2	162.0	176.8			
	Sample 7	159.9	165.2	165.6	169.7	172.0			
	Sample 8	153.2	161.3	169.9	172.3	171.4			
	Sample 9	154.4	163.8	171.4	174.4	173.5			
	Sample 10	164.0	171.5	175.8	179.8	182.8			
Un-mating Force	Sample 1	114.2	120.3	126.2	126.8	127.4	138.6	105.9	122.7
	Sample 2	105.9	113.1	119.8	123.3	124.8			
	Sample 3	106.7	108.4	115.7	119.6	122.3			
	Sample 4	107.6	116.1	122.7	126.1	126.8			
	Sample 5	111.5	112.2	119.4	121.2	122.8			
	Sample 6	110.0	113.2	120.8	126.2	135.9			
	Sample 7	109.7	122.8	129.8	132.0	134.6			
	Sample 8	110.7	124.0	131.6	133.0	134.2			
	Sample 9	114.3	127.9	133.2	138.6	138.2			
	Sample 10	114.3	127.9	133.2	138.6	138.2			

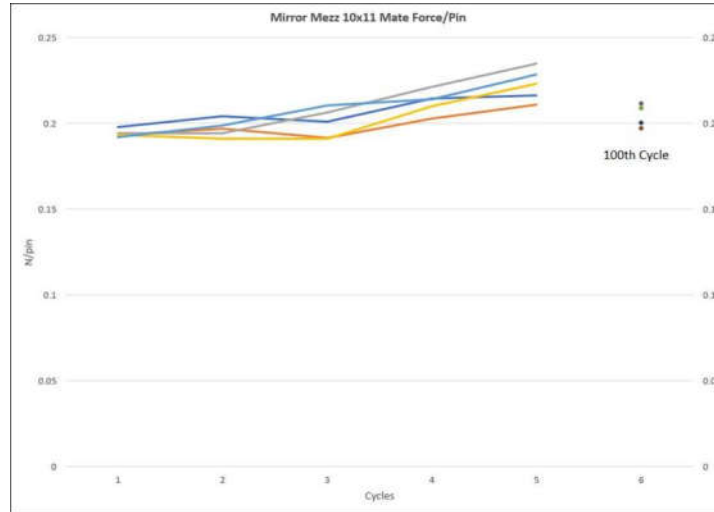


Figure 7 Measured Mate Force per Pin for Molex Mirror Mezz 209311-1115

6.3 Accelerator Module Top Stiffener

The reference model for the accelerator mezzanine module top stiffener is purely reference and dimensions may be changed or adjusted to accommodate the specific application and board layout of the OAM PCB.

6.4 Accelerator Module Bottom Stiffener

The reference model for the accelerator mezzanine module bottom stiffener is shown in Figure 8 Reference Design of Bottom Stiffener. Required dimensions are shown in Figure 9 Bottom Stiffener Required Dimensions. Other features and dimensions of the bottom stiffener shown in the reference model are optional and can be adjusted based on the needs of the module PCB.

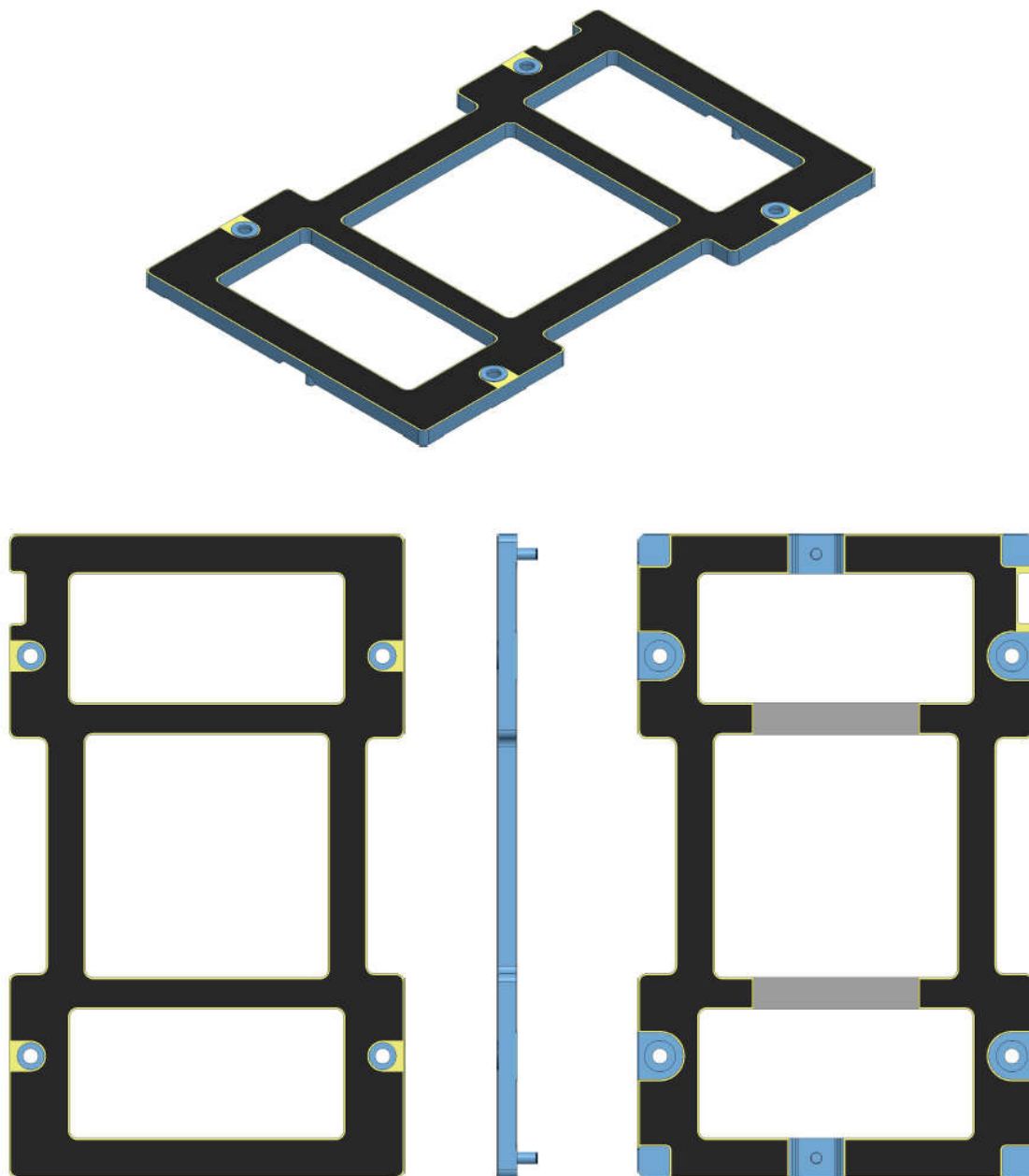


Figure 8 Reference Design of Bottom Stiffener

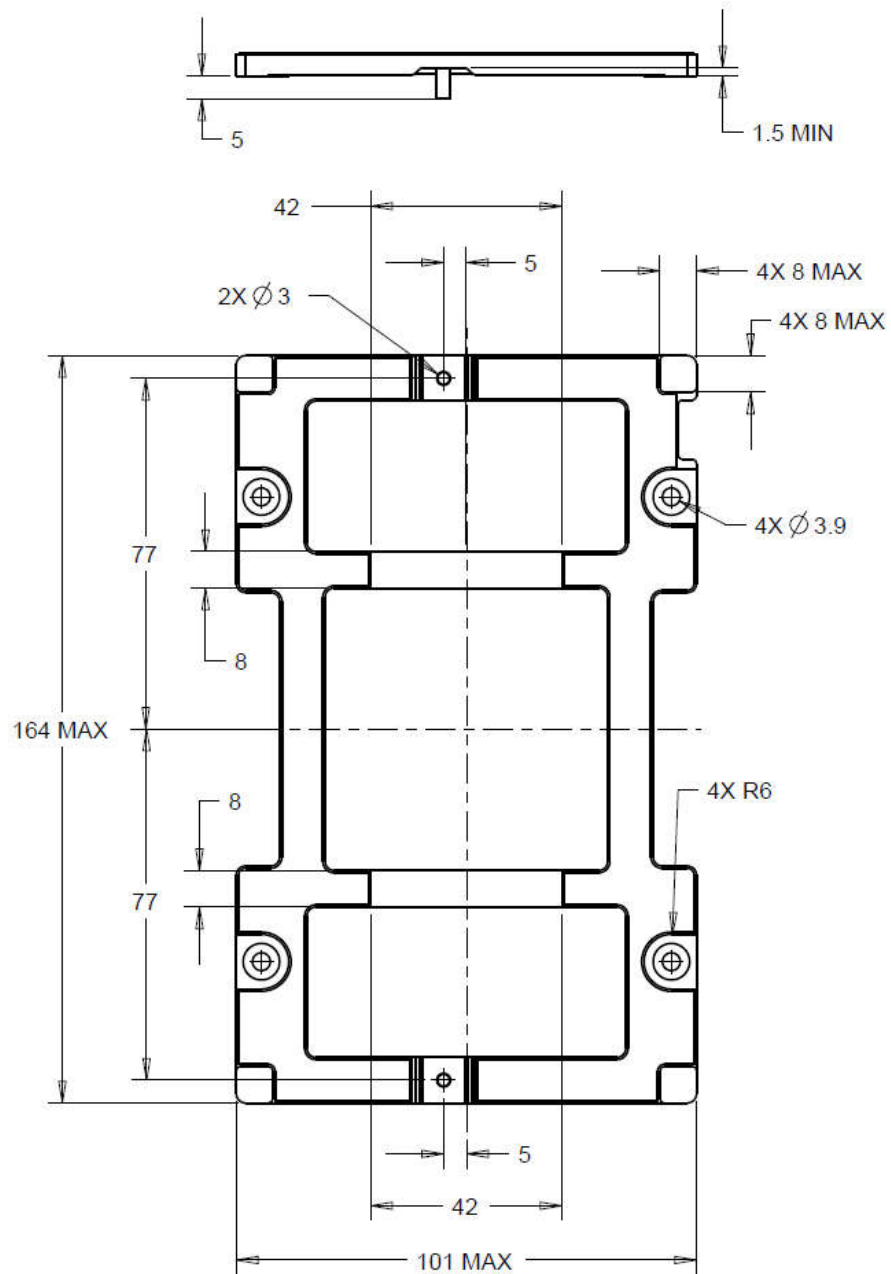


Figure 9 Bottom Stiffener Required Dimensions

6.4.1 Tolerance Stack-up of Bottom Stiffener

Standoff height as recommended by Molex for the Mirror Mezzanine Connector is $5\text{mm} \pm 0.15\text{mm}$. This tolerance may be difficult to attain using an insulator-adhesive-stiffener-adhesive-insulator stack, so it is highly recommended that pockets be machined into the stiffener to account for the tolerances of the insulator and adhesive (see reference design CAD for further details). With a stiffener only stack, 0.15mm should be easily attainable.



Figure 10 Tolerance Stack-up of Bottom Stiffener

6.4.2 Alignment Pins

There are two alignment pins required on the bottom stiffener of the mezzanine module, intended as guidance features as well as an additional keying feature for the module (see Section 6.5 for more details). They are to be defined as 3mm diameter, with a length of 10mm measured from the bottom of the mezzanine PCB. Note that since there may be components on the bottom side of the PCB, if the stiffener is pocketed in this area the total length of the pin will be shorter than 10mm. It is recommended that the minimum thickness of the stiffener is 1mm in these areas. Figure 12 shows an example of a possible alignment pin. Note that the length will vary depending on the specific chosen geometry of the bottom stiffener.

MPN: PEM TPS-3mm-8 or equivalent

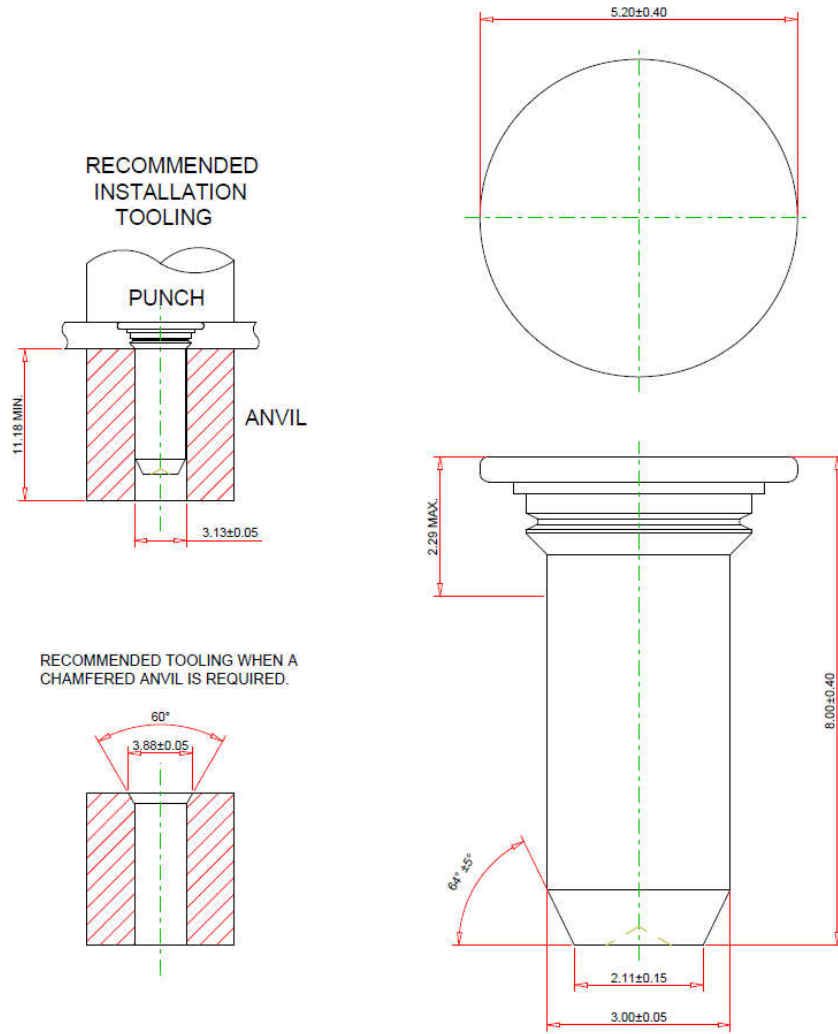


Figure 11 Alignment pin example

6.4.3 EMI Gaskets / Pads

The bottom stiffener has two defined areas of 8x42mm size that are reserved for placement of fabric-over-foam gaskets. This area is designed to have a 0.5mm depth, and the gasket defined should have a 6x40mm footprint, with 1mm height. This provides a 50% nominal compression and solid grounding to the baseboard (which has an equivalently designed ground pad).

MPN: Laird 4Y03PC51H00158 or equivalent

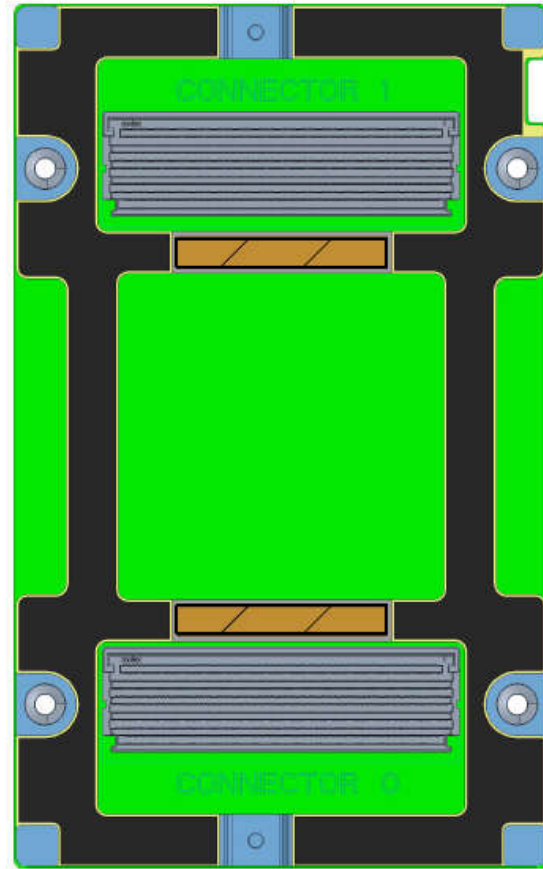


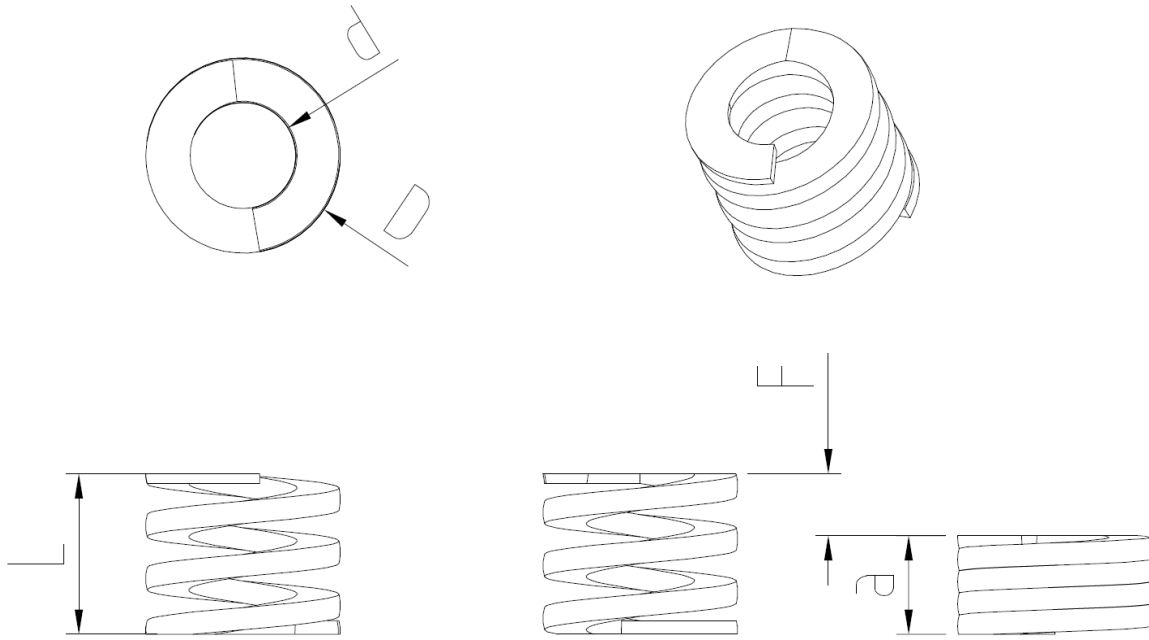
Figure 12 Fabric-over-foam Gasket Locations (brown)

6.4.4 Die Springs

Due to the large number of pins, the mate and unmate forces of the Molex Mirror Mezzanine connectors are high (see Section 6.2.1). To assist with the de-mate, die springs are to be used. It is strongly suggested for OAM vendors to use this reference spring in

However, an equivalent spring shall have a spring constant of at least 70N/mm, and a compression of at least 2.5mm. Inner diameter shall be 4.2mm and outer diameter shall be 7.8mm. These springs fit into 8mm diameter counterbores of 4mm depth in the bottom stiffener. Installation method is using glue (3M DP810 or equivalent), applied with maximum thickness of 0.1mm.

MPN: Timson WG774265 or equivalent



D (mm)	d (mm)	L (mm)	a (mm)	F (mm)	K (N/mm)
$7.8^{+0.05}_{-0.20}$	$4.2^{+0.1}_{-0.15}$	$6.5^{+0.2}_{-0.2}$	$3.6^{+0.15}_{-0.15}$	$2.9^{+0.35}_{-0.35}$	$78.6 \pm 10\%$

Figure 13 Die Spring dimensions and drawing

Table 2 Spring constant and free length of die springs, shown compared to cycle count

	Sample 1		Sample 2		Sample 3		Sample 4		Sample 5	
	L (mm)	K (N/mm)	L (mm)	K (N/mm)	L (mm)	K (N/mm)	L (mm)	K (N/mm)	L (mm)	K (N/mm)
1	6.48	81.87	6.44	80.88	6.48	82.12	6.45	80.20	6.46	80.70
2	6.47	81.23	6.43	79.80	6.47	81.86	6.44	79.98	6.46	80.58
3	6.47	81.01	6.42	79.84	6.47	81.33	6.44	79.96	6.45	80.48
4	6.46	80.95	6.42	79.70	6.46	81.20	6.44	79.80	6.45	80.46
5	6.46	80.95	6.41	79.37	6.46	81.17	6.43	79.68	6.44	80.28
6	6.46	80.90	6.41	79.22	6.46	81.13	6.43	79.40	6.44	80.29
7	6.46	80.79	6.40	79.31	6.45	80.97	6.42	79.52	6.43	80.12
8	6.45	80.77	6.40	79.16	6.45	81.17	6.42	79.48	6.43	80.02
9	6.45	80.76	6.39	79.11	6.43	80.98	6.41	79.47	6.42	79.90
10	6.44	80.68	6.39	79.02	6.43	80.91	6.41	79.38	6.42	79.93
Avg	6.46	80.99	6.41	79.54	6.46	81.28	6.43	79.69	6.44	80.28

6.5 Baseboard Keepout Zone & Grounding Pads

The below figure shows the baseboard outline (top side view) required to accommodate this module. All cross-hatched areas are required to be grounded except for the four corner 10x10 square holes. The 10x10 square holes in the corners are highly recommended to be grounded. Additionally, it is recommended to route high-speed traces away from mounting hole areas due to large compression forces from the die spring.

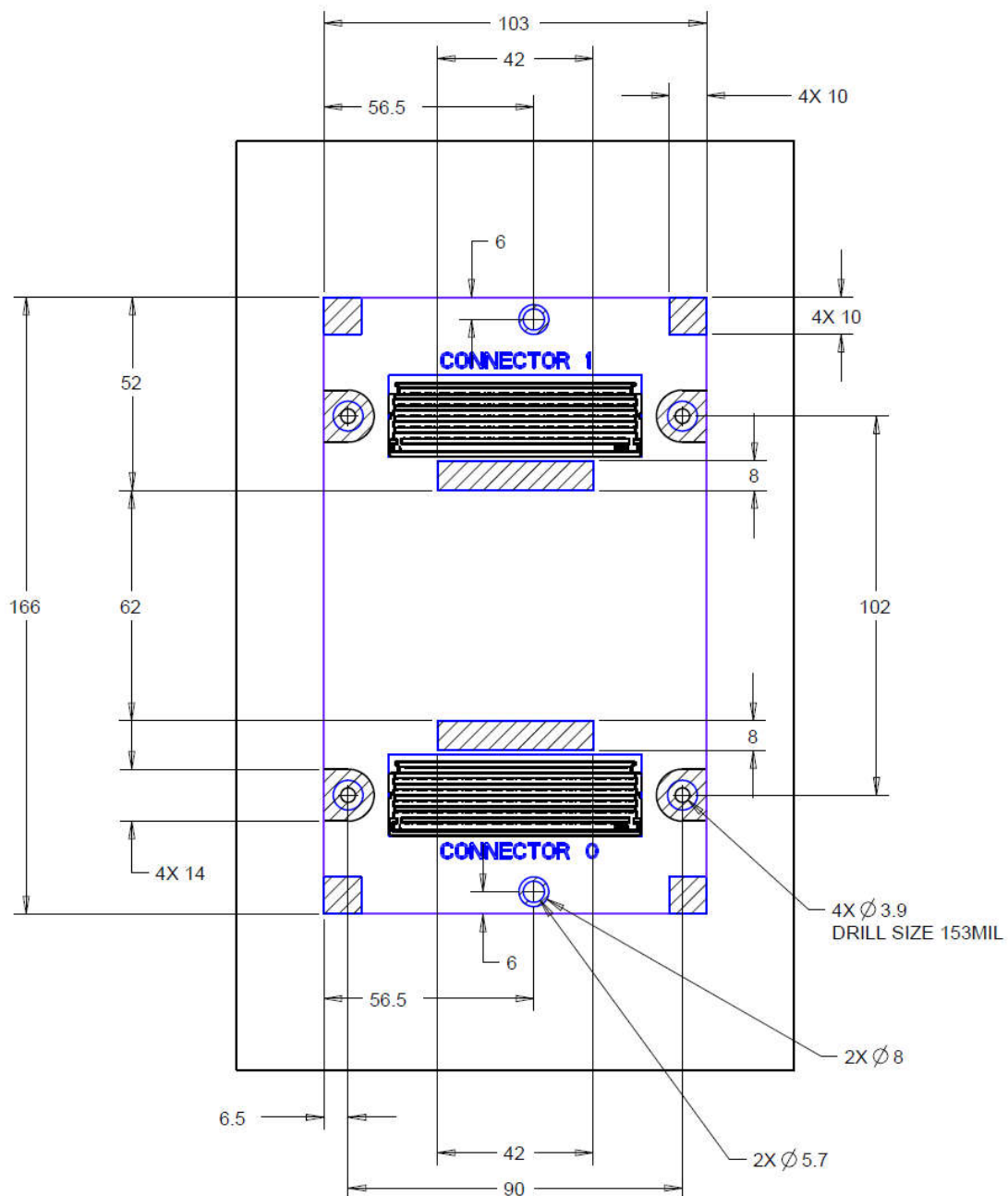


Figure 14 Baseboard KOZ and Grounding Pad Dimensions

6.5.1 SMT Nut

Two SMT nuts with the dimensions shown in Figure 17 are to be soldered to the baseboard in the locations with 5.7mm diameter holes. These nuts provide the mating features to the alignment pins on the bottom stiffener of the mezzanine module. Clearance of the 3mm pins in the 3.6mm nuts means that the module will come within 0.3mm of its final position.

MPN: Ray Home 1000401319 or equivalent

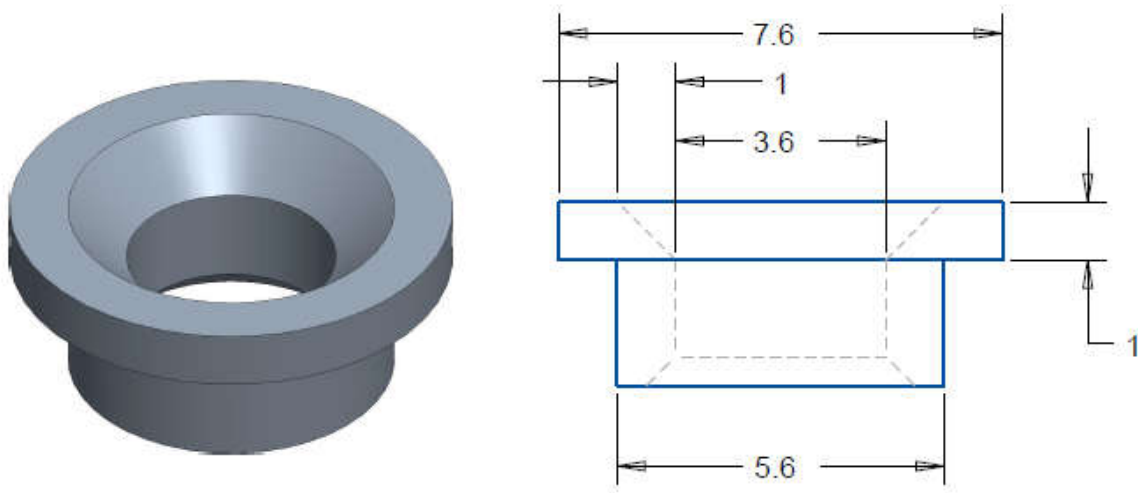


Figure 15 SMT Receiving Nut for Baseboard

6.5.2 Component Keep-out Zone

The baseboard has a component keep-out zone of 103x166mm, as shown in Figure 14 Baseboard KOZ and Grounding Pad Dimensions.

6.5.3 Grounding Pads

As with the bottom stiffener, the baseboard has two grounding pads of size 8x42mm, for the EMI fabric-over-foam gaskets on the stiffener to provide good contact. Refer to Section 4.3.3 for gasket MPN and description.

6.6 Recommended Alignment Features

There are three stages of engagement when installing the mezzanine module to system.

Stage 1: Notch in top of heatsink providing visual guidance and orientation reference. Reference design is shown with 1mm clearances (plastic top is 103mm with a 0.5mm bumper on each side of the module).

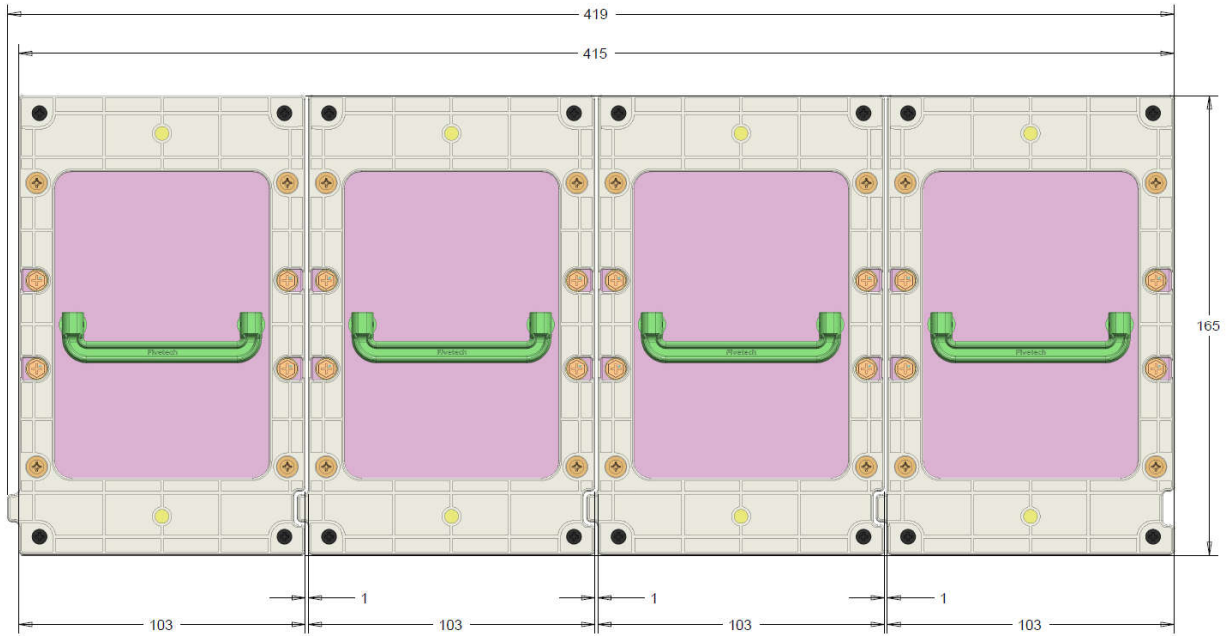


Figure 16 Top view of four adjacent mezzanine modules with heatsinks

Stage 2: Alignment pins, two 3mm pins from the OAM into two 3.6mm SMT nuts on baseboard.



Figure 17 Side view (exploded) showing alignment pins being received by 1mm tall SMT nuts

Stage 3: Connector housing built-in engagement (Molex Mirror Mezz gatherability: 0.76mm).



Figure 18 Side view (exploded) showing mezzanine connectors doing final alignment

6.7 Reference Heatsink Design

It is recommended to use an air-cooled solution for TDP equal or less than 450W modules. For modules which are over 450W, it is recommended to consider other solutions such as liquid cooling.

The below figure shows the reference model of heatsink with accelerator module assembly.

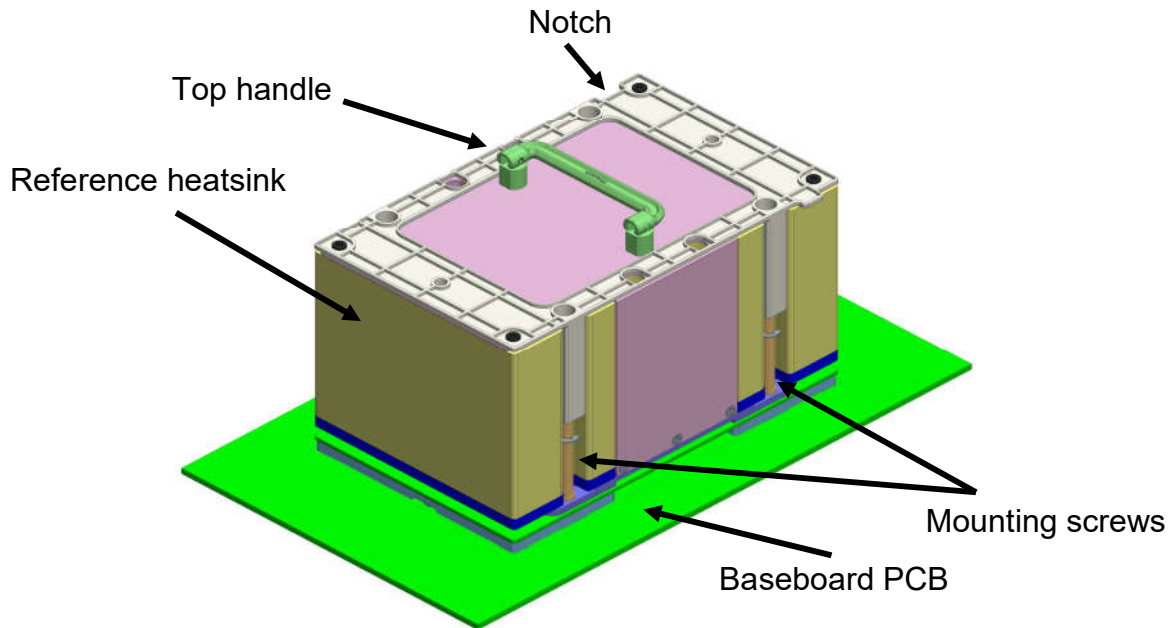


Figure 19 Reference Model of Heatsink with Accelerator Module Assembly

6.7.1 Top Handle

Due to the size and bulk of the heatsink and module assembly, a handle is recommended. The reference design uses a folding handle. This handle is screwed into a sheet metal panel which is then attached to the heat sink base with six M2.5 flathead screws. This method of attachment allows the load to be transferred through the more rugged base instead of through the delicate heat sink fins.

MPN: Fivetech 62-57P-064-7-02-5 or equivalent

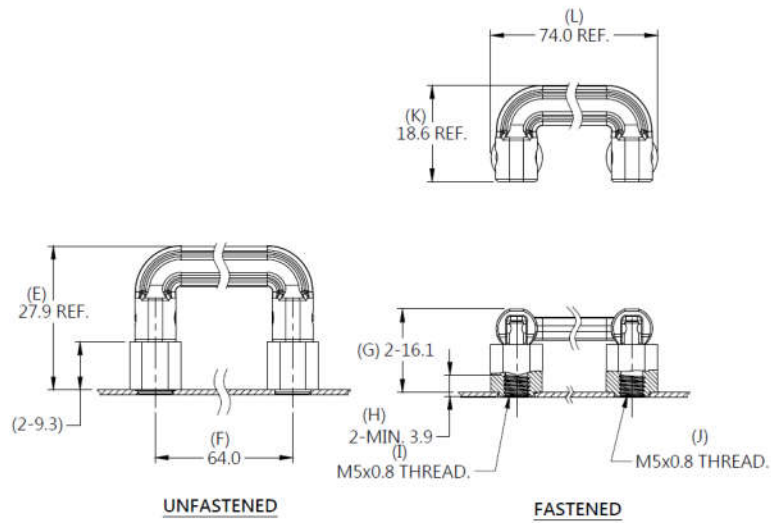


Figure 20 Dimensioned Smart Folding Handle from Reference Design

6.7.2 Long Screw Attachment

A set of four M3.5, spring-loaded, Phillips head long screws are used to attach the module to the baseboard. Note that the reference screw provided is simply a reference and that L3 and L4 will need to be adjusted based on thicknesses of the baseboard and bolster plates. However, these mounting screw locations are fixed per the requirements of the mezzanine board layout and the baseboard layout. Each screw clears the top stiffener, mezzanine PCB, and bottom stiffener (including the die spring), and the baseboard to screw directly into the bolster plate below the baseboard PCB. It is recommended that the OAM is attached to the baseboard by torqueing the screws in a diagonal pattern.

MPN of Long Screw: Wujiang Screw MDCM0359733N or similar

MPN of Spring for Long Screw: Surpassing Hardware Spring FDJG7004010 or similar

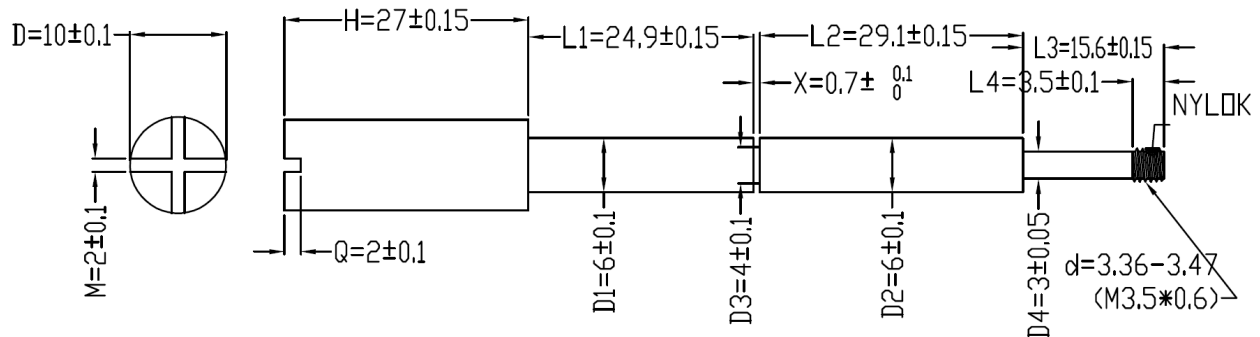


Figure 21 Drawing of Wujiang Screw MDCM0359733N

7 Thermal Specification

7.1 Environmental Conditions

To meet the thermal reliability requirement, the thermal and cooling solution should dissipate heat from the components when the module is operating at its thermal design power. The module should be able to operate in the following environmental conditions without any throttling or thermal issues:

- Ambient temperature: 5°C to 35 °C
- Approach temperature: 10°C to 48 °C, considering shadowing other components
- Altitude: sea level to 3000 ft*, without temperature deration
- Relative Humidity: 20% to 90%
- Cold boot temperature: module should be able to boot and operate at an initial temperature of 10°C

*An extended altitude range of up to 6000ft is recommended.

In addition, the module should be able to remain unaffected at non-operational storage temperature range of -20°C to 85°C.

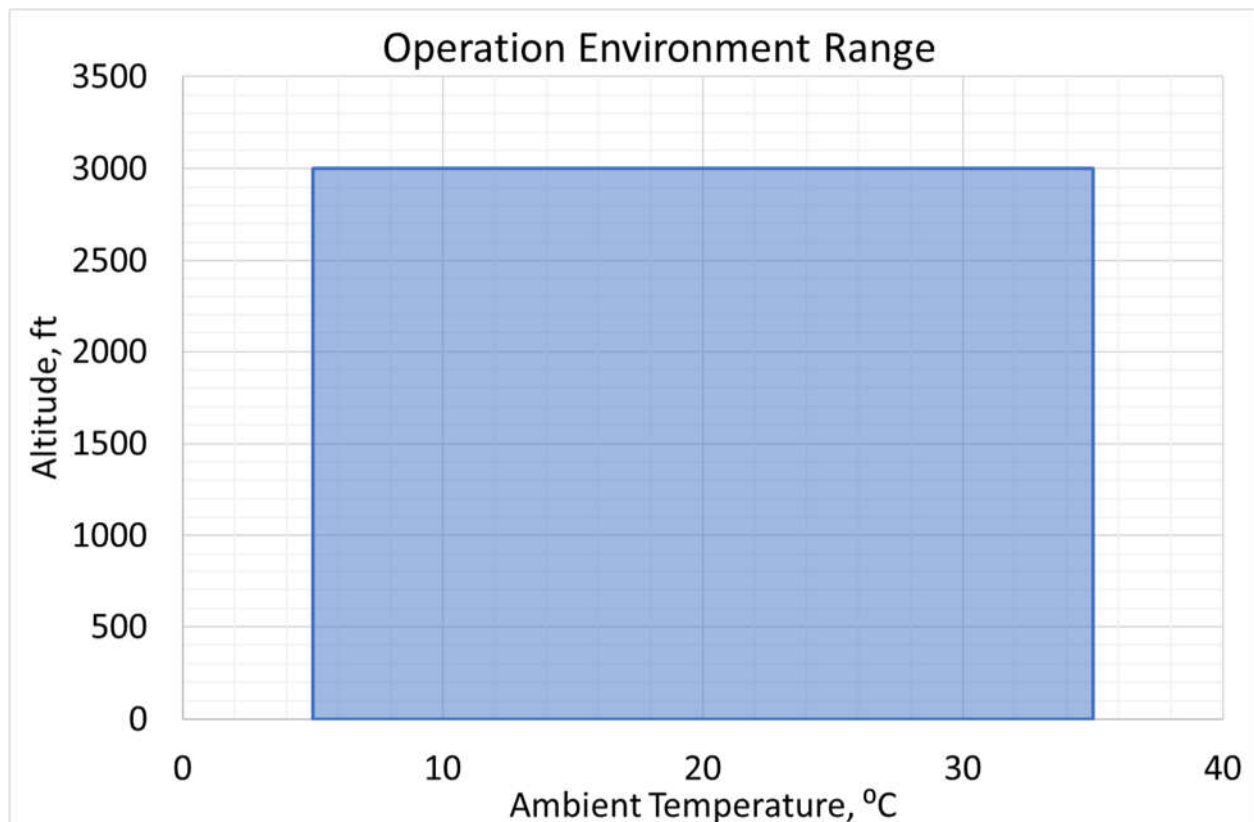


Figure 22 Module Operation Ambient Temperature

7.2 Temperature Report

7.2.1 Temperature Sensors

The module reports readings of ASIC temperature sensor and HBM temperature sensor to support software or hardware throttling, shutdown, and drive fan speed through BMC. The sensors should be located or calibrated to:

- Always report the hottest junction temperature in the component
- Keep accuracy within $\pm 3^{\circ}\text{C}$

Lower temperature limit, non-critical temperature limit, and critical temperature limit should be defined for those temperature sensors to support throttling or shutdown features.

7.2.2 Remaining Components

For the remaining components that are not monitored by temperature sensors or not included in fan speed control (FSC), their cooling solutions should be properly designed such that:

- Before ASIC or Memory temperature readings reach throttling thresholds, they will be maintained below the temperature limits.
- When any ASIC or Memory temperature reading reaches a throttling threshold but not the hardware shutdown limit, these components will remain functional to support reduced functionality of the module.

7.3 Thermal module info

To enable the module with appropriate cooling solutions, supplier will provide the following thermal info for each product model:

- ASIC & Memory (HBM or DRAM) junction temperature limit
- ASIC & Memory (HBM or DRAM) junction to surface/case temperature correlations
- Connector surface temperature limit
- ASIC & Memory (HBM or DRAM) junction temperature range at nominal operation conditions

7.4 Heatsink Assembly

To minimize complexity of assembly, servicing and risk of failure, the module will meet these requirements:

- Only one replaceable heatsink assembly (primary heatsink) is needed for the module, which can be swapped in field.
- The other heatsink parts (i.e. secondary heatsinks) and thermal interface materials will come with the module, and do not need replacement over the module lifetime.

Reliability test reports will be provided to validate lifetime of the thermal interface materials. Shock and Vibration test reports will be provided to validate robustness of the module assembly.

7.5 Thermal Recommendation

7.5.1 Airflow Budget

Considering the limit on air delivery/removal capabilities of typical infrastructures, it is recommended that the OAM module be capable of operating with full performance at or below an airflow/power ratio of 0.145 CFM/W, with ambient temperature up to 30°C at sea level. This is equivalent to an inlet/outlet air temperature increase of 22°F.

- For operation at altitude, the same air temperature difference of 22°F is recommended.
- For a single OAM that is shadowed by other components, the airflow/power ratio is calculated with airflow through its heatsink, and the module power
- For an OAM shadowing other components or multiple OAMs in serial, this calculation uses the airflow through the flow channel, and the sum of the power of OAM modules and upstream components.
- For OAM modules with power lower than 300W, an airflow/power ratio of 0.1 CFM/W or lower is usually achievable and recommended.

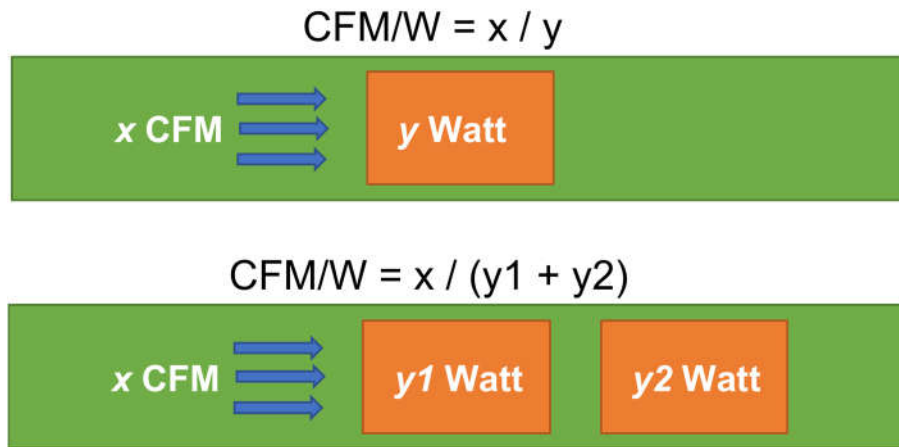


Figure 23 CFM per Watt

7.5.2 Reference Heatsink Design

Please refer to Figure 19 Reference Model of Heatsink with Accelerator Module Assembly. To help the enablement of each product, a reference heatsink design will be provided, including

- Thermal simulation model
- 3D mechanical drawing

Performance of the reference heatsink is provided in Figure 23, the thermal resistance of which is calculated based on:

$$R_{ca} = \frac{T_{case} - T_{LA}}{P_{die}},$$

Where T_{case} is the surface center temp of heater, T_{LA} is the approaching temperature and P_{die} is the power of the heater indicating the die instead of the total module power.

Die size and power density plays an important role in the thermal performance of OAM module. As a general guidance, this chart provides curves of three different die (heater) sizes. Each product can make preliminary estimation by referring to the curve with closest size.

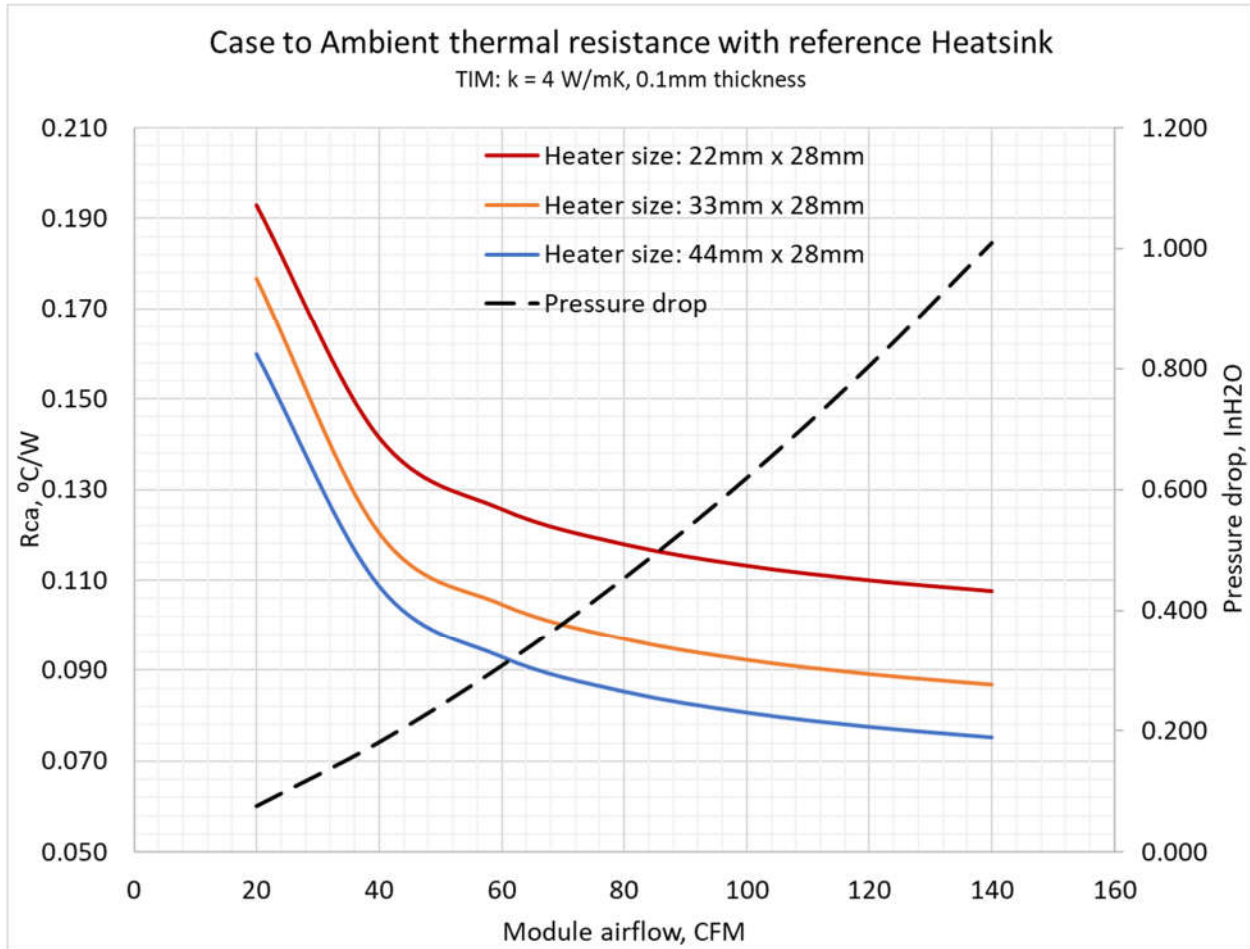


Figure 24 Thermal resistance and pressure drop of reference heatsink

If applicable, significant improvement can be achieved by implementing vapor chamber to assist heat spreading in the base. The performance of Reference heatsink design V2 with vapor chamber base is provided as follows:

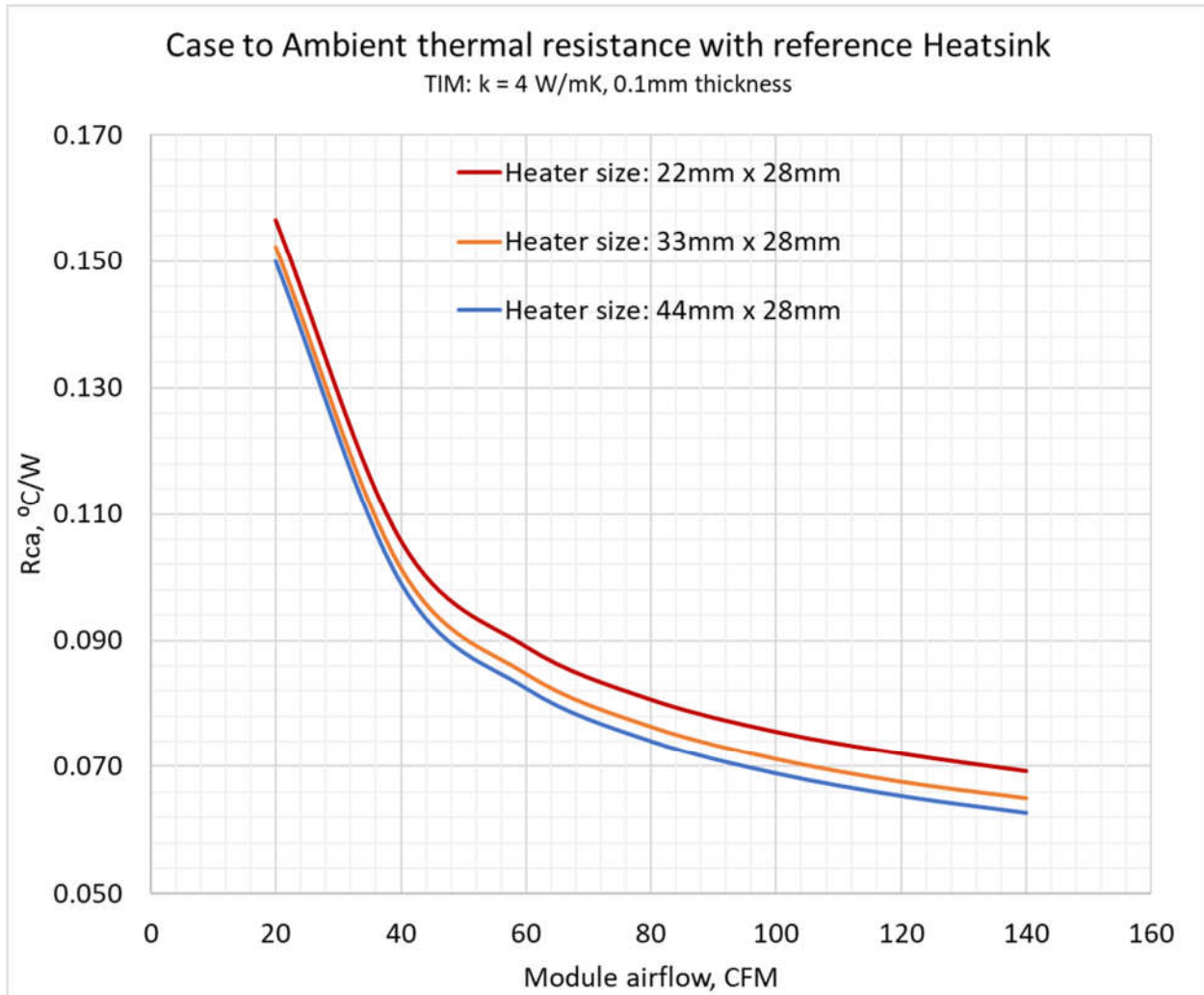


Figure 25 Thermal resistance of reference heatsink V2 with vapor chamber

7.5.3 Cooling Limit

Depending on model and application, the OAM may operate at a variety of power levels. However, traditional air-cooled heatsinks may hit their performance limit due to the constraint on heat spreading technologies. Beyond a certain chassis height, fin size, and airflow rate, the improvement on thermal resistance of air-cooled heatsink becomes minimal.

Package size also have significant impact on the cooling capability of OAM modules. Figure 26 provides the airflow needs of single OAM module at given approaching temperature, case temperature target, thermal interface material and die powers. Beyond 120CFM more airflow towards OAM brings diminishing return, which limits the max OAM power supported. This can be also used to estimate cooling capability of system design and fan trays.

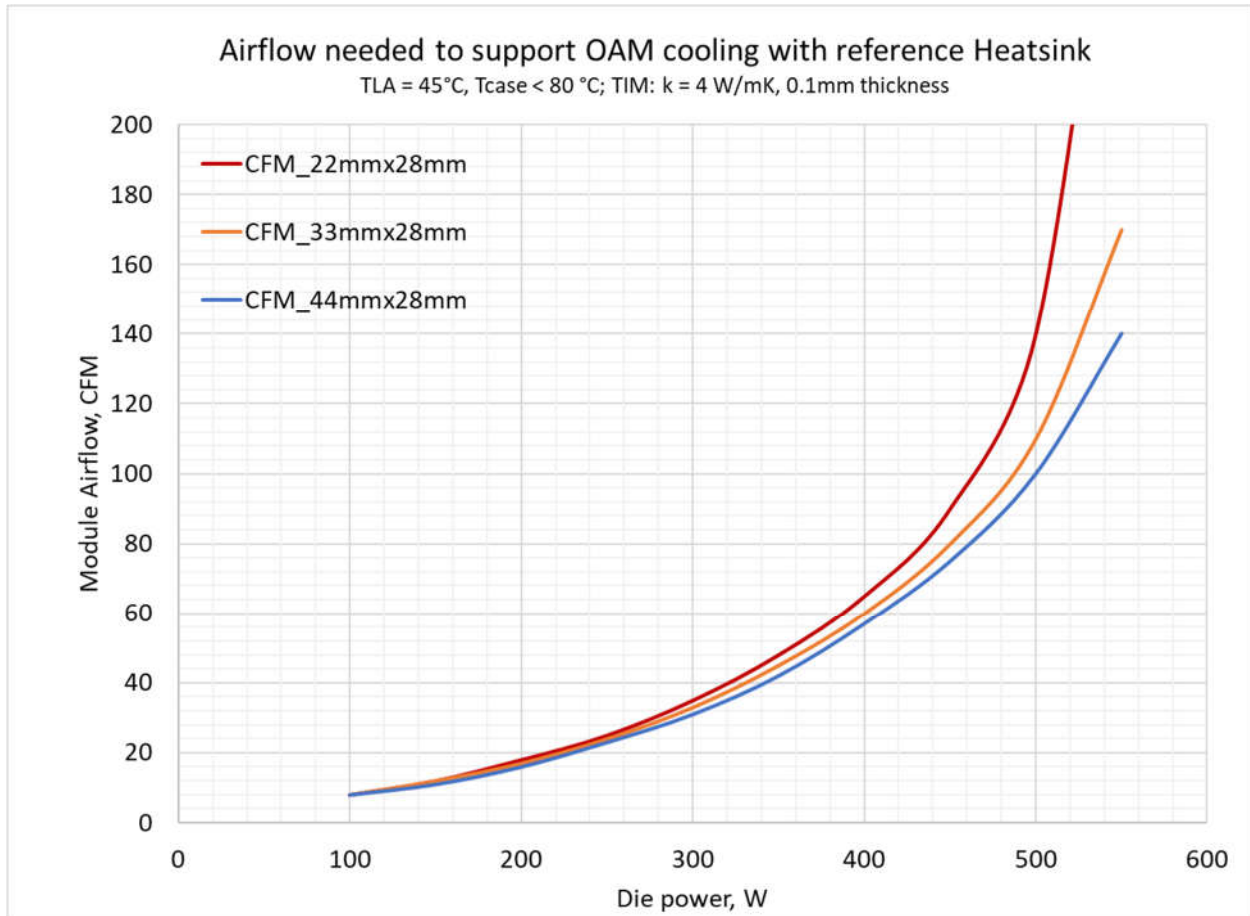


Figure 26 Increasing need of airflow for OAM cooling as die power increases

For a reference OAM in a typical platform with 8x OAMs, shadowing layout, it is observed that the maximum module power that air cooling can support is approximately 450W. Beyond this power limit, advanced cooling solutions are recommended to support its operation at the hotter part of the operational boundary condition range. These advanced cooling solutions would also be recommended for extended environment boundary conditions. Note that this limit may vary for different products, depending on die size, power distribution, and junction temperature limits.

Open loop liquid cooling is one of the feasible cooling solutions to support modules of higher power. To support typical open loop liquid cooling modules designed for a 1RU (height = 44.45mm) system, it is recommended that OAM vendors limit the maximum distance from the lower surface of bottom stiffener to the top surface of the die (ASIC/HBM) to within 13mm.



Figure 27 Maximum height of OAM to enable liquid cooling within a 1RU system

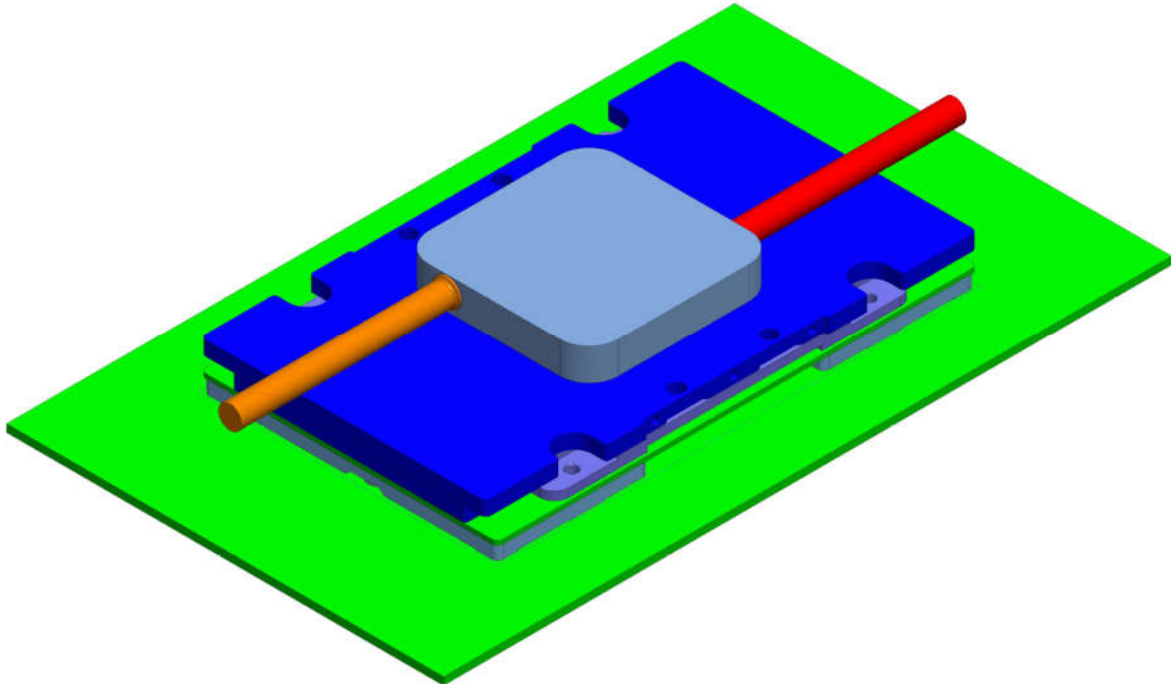


Figure 28 An Example of Open Loop Liquid Cooling setup concept for OAM

A typical open loop liquid cooling setup (cold plate) for the OAM may include the following parts:

- Cold plate base + thermal interface materials
- Internal Mini/Micro channels
- Internal Manifold
- Coolant inlet/outlet tubes

With a proper coolant supply, open loop liquid cooling has the potential of delivering surface-to-coolant thermal resistance lower than $0.05^{\circ}\text{C}/\text{W}$. However, it would require liquid supply and control systems to be established as part of the data center infrastructure.

7.5.4 Heatsink Installation

A lot of OAM modules use a bare die design, which may be fragile and susceptible to imbalance of pressure on its surface. The system integrator should contact the accelerator module supplier for the maximum static and dynamic pressure for the die, to guide installation of the primary heatsink to the

module. The static mounting pressure should also be high enough to enable optimum performance of the TIM material.

We suggested that following guidelines to be followed during installation:

- Screw head type: Philips #2
- Tightening pattern: Diagonal
- Tightening stage: multiple stages, 2 or 3
- Tightening torques: (TBD)

The mounting pressure of heatsink is determined by:

- Max pressure the package can sustain
- Min pressure the TIM need to deliver enough performance

We recommend the mounting pressure range to be 30 ~ 60 psi for OAM with bare die packages. For engineering samples without enough assembly yield rate learnings, we recommend starting with an initial mounting pressure of 15~30 psi. For lid-covered OAM packages, the mounting pressure is yet to be explored.

7.5.5 Thermal Interface Material

The thermal interface material between the die (ASIC/HBM) and the primary heatsink should maintain a thermal conductivity of at least 4W/m·K through the end of its life. This is equivalent to approximately 6°C temperature difference between the heatsink base and the die top surface, for a bond line thickness of 0.1mm and heat flux of 24.4W/cm², which is also equivalent to 300W uniformly distributed over a surface area of 44mm x 28mm.

Maximum warpage of the package should not exceed 0.2mm. This could potentially lead to an average bond line thickness of 0.1mm for the TIM. Varying for different die sizes, TIM could easily contribute 0.01 °C/W ~ 0.08 °C/W, up to 50% of total thermal resistance:

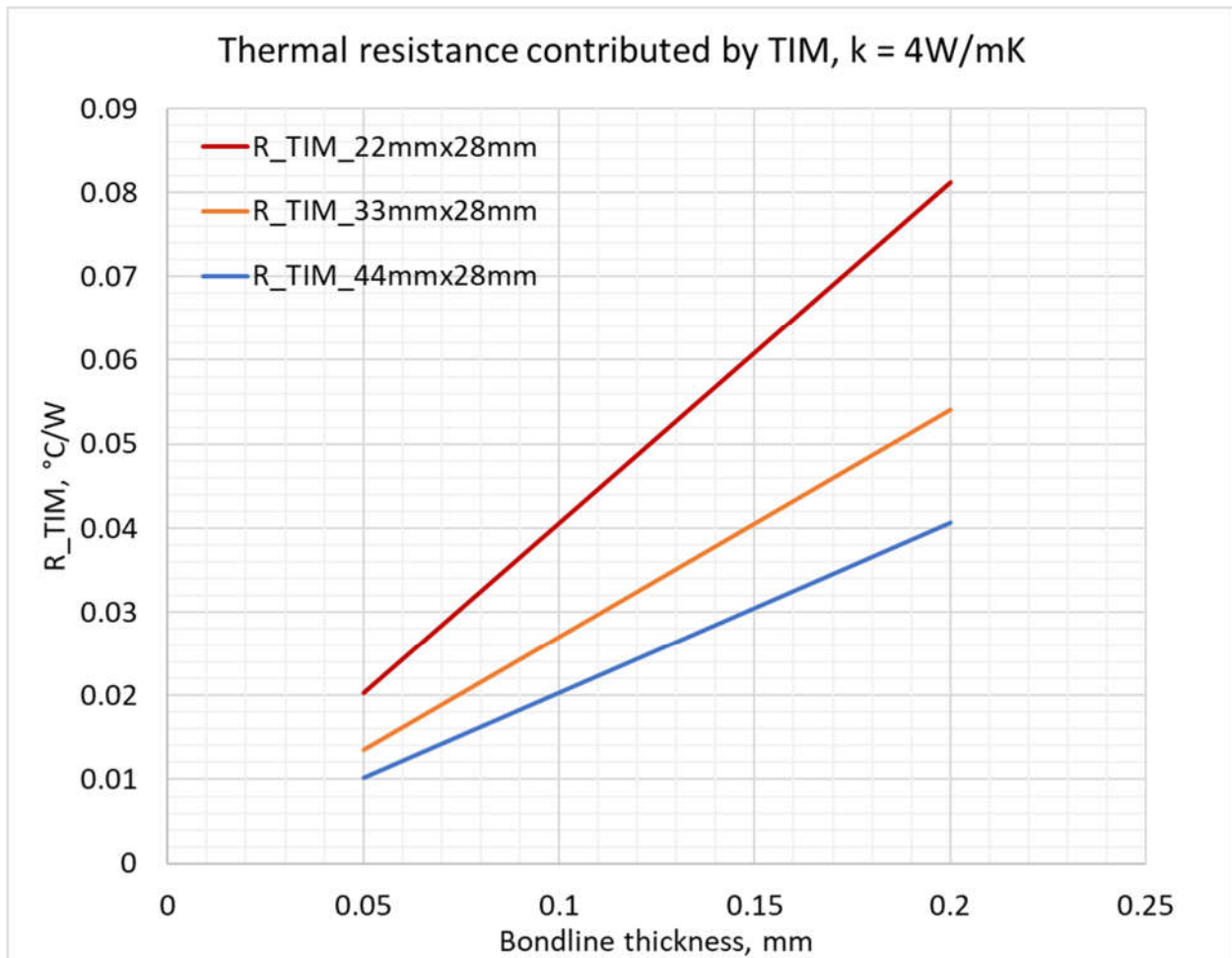


Figure 29 Thermal resistance across TIM layer at different bond line thicknesses

8 Accelerator Module Electrical Specification

8.1 Electrical Connector

The module utilizes two 688pin Molex Mirror Mezz connectors. It is a BGA attached connector and supports bit rates up to 56Gbps NRZ or 112 Gbps PAM4 in a 90 Ohms nominal impedance $\pm 5\%$ tolerance which make it compatible to support typical 85 Ohms based interfaces such as PCIe Gen3/4/5 as well as other 100 Ohms based high speed interfaces. All power and I/O signals are routed through the two connectors down to the system baseboard. The system baseboard should connect these signals to the appropriate circuitry depending on the required feature sets. The below table lists the electrical requirements for the module connectors.

Table 3 Electrical Requirements for Molex Mirror Mezz

Items	Mirror Mezz
Data Rate Support	25/28/32/56Gbps NRZ , 56G/112G* PAM4
Connector Impedance	90ohm $\pm 5\%$
Differential pairs per two connectors	172 pairs
Pin Pitch	0.9mm and 1.3mm
Current Rating per pin @80C ambient temp, 1.5oz copper	1A/pin after 20% derating
Max Voltage Application	30V AC (OAM supports 60V after Molex's pin assignment review)
Connector insertion cycles	100cycles
Withstand voltage	500V min
Low Level Contact Resistance (max initial):	30m Ω for 5mm stack height
Insulation resistance	1-M Ω min
Intra-pair skew	≤ 5 ps

*note:

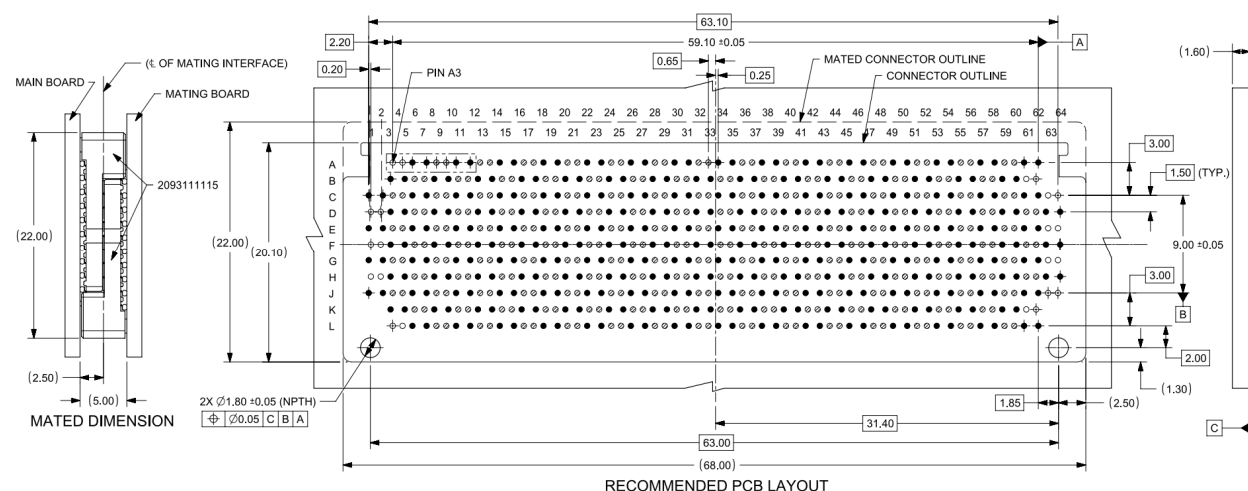


Figure 30 Mirror Mezz Connector Footprint

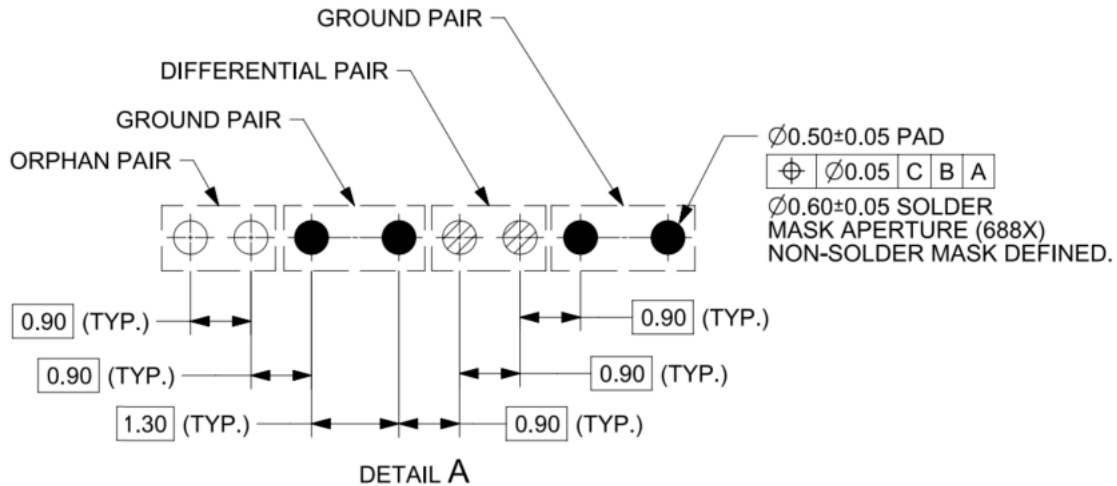


Figure 31 Mirror Mezz Connector Pin to Pin Pitch

8.2 Mezzanine Module Connector Pinout Quadrants

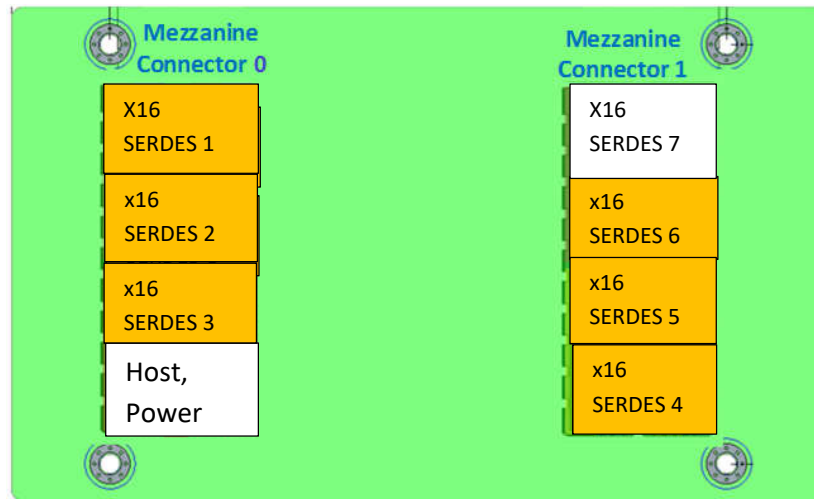


Figure 32 Mezzanine Connectors Pinout Quadrants

The OAM Connector 0 has the following interfaces:

- 54V/48V and 12V input power
- x16 SerDes to connect to host
- 3 x16 SerDes for accelerator to accelerator communication
 - X16 may be split to sub links like 2* x8s or 4* x4s or 16*x1s.
 - If the ASIC or ASICs on the module only support x8 or x4 per SerDes, it should start from Lane0 from the SerDes, e.g. lane [7:0] or lane [3:0].
 - We do not recommend lane reversal support on the baseboard due to modules having the option to be 1X16 link or 2X8 or 4X4 links.

- Other single ended signals like PRESENT#, I2C, GPIOs etc.

The OAM Connector 1 has the following interfaces:

- Power pins for 3.3V
- Other single ended signals like JTAG, GPIOs etc.
- Up to 4 SerDes for accelerator to accelerator communication or other purposes:
 - SerDes 4, 5, 6 and 7 are up to x16 lanes which can be split to X8s, X4s or X1s.
- SerDes 7 may be defined for different use cases:
 - This link could be the 7th SerDes for some cases to have fully connected interconnect between the modules
 - It could be the 2nd link to host for the ASIC(s) on the module, e.g. a full x16 link, 2 x8, or 4 x4 links.
 - Or it could be special defined link by some ASICs. E.g., it could be a downstream port for the ASIC on the module.

8.3 Accelerator Module Pinout Description

The detail pin mapping to connectors will be provided in separated spreadsheet. This section only shows the pin list and description.

Table 4 Accelerator Module Pinouts

Signal	Module Direction POV	Description	Voltage	Required or Optional	Total Diff Pins	Total Single Pins	Conn 0 or 1
P48V	Power Input	44V-59.5V main voltage for high power applications. Up to 700W when $V_{in} \geq 44V$. The module should be able to operate at 40V to 44V but at lower power (motherboard to drive PWRBRK# for $V_{in} < 44V$).	44V-59.5V	Required		16	Conn0
P12V1	Power Input	12V mandatory Board Infrastructure Power. Up to 50W	12V	Required		5	Conn0
P12V2	Power Input	12V main Power for low power applications. Up to 300W. For 12V motherboard/module designs, P12V1 and P12V2 can be shorted	12V	Required for P12V based OAM		27	Conn0

		together for up to 350W combined power					
P3V3	Power Input	3.3V Main Power. Up to 5W	3.3V	Required		2	Conn0
PVREF	Power Output	Low voltage output for GPU/ASIC sideband I/O reference on motherboard components. Module should provision maximum 0.5A to be provided to the motherboard. V_{ref} can be any value between 1.5V to 3.3V set as per GPU/ASIC sideband I/O voltage.	V_{ref}	Required		2	Conn0
PETp/n [15:0]	Output	PCIe or equivalent host link Transmit differential pairs. Module Transmit, Host Receive. Note: AC coupling caps must be placed on the motherboard side.		Required	16		Conn0
PERp/n [15:0]	Input	PCIe or equivalent host link Receive differential pairs. Module Receive, Host Transmit. Note: AC coupling caps must be placed on the motherboard side.		Required	16		Conn0
SERDES_1 Tp/n [15:0]	Output	SerDes link 1 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn0
SERDES_1 Rp/n [15:0]	Input	SerDes link 1 Receive differential pairs.		Required	16		Conn0
SERDES_2 Tp/n [15:0]	Output	SerDes link 2 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn0

SERDES_2 Rp/n [15:0]	Input	SerDes link 2 Receive differential pairs.		Required	16		Conn0
SERDES_3 Tp/n[15:0]	Output	SerDes link 3 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn0
SERDES_3 Rp/n [15:0]	Input	SerDes link 3 Receive differential pairs.		Required	16		Conn0
SERDES_4 Tp/n[15:0]	Output	SerDes link 4 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn1
SERDES_4 Rp/n [15:0]	Input	SerDes link 4 Receive differential pairs.		Required	16		Conn1
SERDES_5 Tp/n [15:0]	Output	SerDes link 5 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn1
SERDES_5 Rp/n [15:0]	Input	SerDes link 5 Receive differential pairs.		Required	16		Conn1
SERDES_6 Tp/n [15:0]	Output	SerDes link 6 Transmit differential pairs. AC caps must be placed on Module/die (if required)		Required	16		Conn1
SERDES_6 Rp/n [15:0]	Input	SerDes link 6 Receive differential pairs.		Required	16		Conn1
SERDES_7 Tp/n [15:0]	Output	SerDes Reserved Link Transmit differential pairs. Interconnect requirements defined by Module and System integrator. AC caps must be placed on the Motherboard (if Required)		Optional	16		Conn1
SERDES_7 Rp/n [15:0]	Input	SerDes Reserved Link Receive differential pairs. Interconnect		Optional	16		Conn1

		requirements defined by Module and System integrator. AC caps must be placed on the Motherboard (if Required)					
PE_REFCLKp/n	Input	PCIe Reference Clock. 100MHz PCIe Gen 5 compliant.		Required	1		Conn0
AUX_100M_REFCLKp/n	Input	100MHz PCIe Gen 5 compliant Auxiliary Reference Clock.		Required	1		Conn1
DWN_REFCLKp/n	Output	Downstream Reference Clock. Vendor specific.		Optional	1		Conn1
AUX_156M_REFCLKp/n	Input	156.25MHz TBD compliant Auxiliary Reference Clock		Required	1		Conn1
PERST#	Input	CEM Compliant PCIe Reset	3.3V	Required		1	Conn0
WARMRST#	Input	Warm Reset	V _{ref}	Optional		1	Conn0
DWN_PERST#	Output	Down device PCIe Reset. Vendor specific.	3.3V	Optional		1	Conn1
HOST_PWRGD	Input	Host power good. Active high when P48V, P12V1/P12V2, P3V3 voltages are stable and within specifications. This is considered the “Power Enable” signal for the module.	3.3V	Required		1	Conn0
MODULE_PWRGD	Output	Module power good. Active high when the module has completed its own power up sequence and is ready for PERST# de-assertion	3.3V	Required		1	Conn0
PWRBRK#	Input	Emergency power reduction. CEM Compliant Power Break	3.3V	Required		1	Conn0
PWRRDT# [1:0]	Input	Power Reduction GPIO to instruct Oam to go certain stage to reduce power 11 - default state L0, normal power	3.3V	Required		2	Conn1

		10 - L1, 1st level power reduction. 01 - L2, 2nd level power reduction. 00 - L3, max power reduction. Details defined by specific OAM product specification.					
THERMTR IP#	Output	Catastrophic thermal event for module components. Active low and latched by the Module logic. Released when the motherboard power cycles the module input voltages	3.3V	Required		1	Conn1
MODULE_ID[4:0]	Input	Module node identifier (e.g. Module #0, #1,...#n). Module has weak PU to drive to 1 by default.	GND	Required		5	Conn0
LINK_CONFIG[4:0]	Input	Mezz Module Host Interface/SerDes Link Configuration and topology. See link config table for details. Module has weak PU to Vref to drive high by default.	GND	Required		5	Conn1
PE_BIF[1:0]	Output	x16 Host Interface Bifurcation Configuration. This output of the module informs the host if it needs to bifurcate the PCIe interface to the module. 00 = one x16 PCIe host interface 01 = bifurcation into two x8 PCIe host interfaces 10 = bifurcation into four x4 PCIe host interfaces 11 = reserved"	V _{ref}	Required		2	Conn1

		Tied to GND on module for logic 0, leave open on module for logic 1; pull up on baseboard					
PLINK_CAP	Output	"P" Port Module Capability support: '0' = PCIe only support '1' = Alternate protocol supported The host system requests an alternate host link protocol by pulling up LINK_CONFIG[0] and the Module informs the system of protocol support on the "P" link via this pin. If the module only supports PCIe as host, this signal is PD on the module.	V _{ref}	Required		1	Conn1
I2C_SLV_D	Bi-directional	Slave I2C data	3.3V	Required		1	Conn0
I2C_SLV_CLK	Input	Slave I2C clock	3.3V	Required		1	Conn0
I2C_SLV_ALERT#	Output	Slave I2C alert indication	3.3V	Required		1	Conn0
I2C_D	Bi-directional	Master I2C/SMBus data. PU on OAM.	V _{ref}	Required		1	Conn0
I2C_CLK	Output	Master I2C/SMBus clock. PU on OAM.	V _{ref}	Required		1	Conn0
UART_TXD	Output	Serial Port Transmit	3.3V	Required		1	Conn0
UART_RXD	Input	Serial Port Receive	3.3V	Required		1	Conn0
JTAG0_TRST	Input	Low Voltage ASIC/GPU JTAG Test Reset	V _{ref}	Required		1	Conn0
JTAG0_TMS	Input	Low Voltage ASIC/GPU JTAG Test Mode Select	V _{ref}	Required		1	Conn0
JTAG0_TCK	Input	Low Voltage ASIC/GPU JTAG Test Clock	V _{ref}	Required		1	Conn0
JTAG0_TDO	Output	Low Voltage ASIC/GPU JTAG Test Output	V _{ref}	Required		1	Conn0
JTAG0_TDI	Input	Low Voltage ASIC/GPU JTAG Test Input	V _{ref}	Required		1	Conn0

JTAG1_TRST	Input	High Voltage JTAG Test Reset	3.3V	Optional		1	Conn1
JTAG1_TMS	Input	High Voltage JTAG Test Mode Select	3.3V	Optional		1	Conn1
JTAG1_TCK	Input	High Voltage JTAG Test Clock	3.3V	Optional		1	Conn1
JTAG1_TDO	Output	High Voltage JTAG Test Output	3.3V	Optional		1	Conn1
JTAG1_TDI	Input	High Voltage JTAG Test Input	3.3V	Optional		1	Conn1
CONN1_I_NITMODE	Output	QSFP-DD Connector 1 Module Initialization mode	V _{ref}	Optional		1	Conn1
CONN1_I_NT#	Input	QSFP-DD Connector 1 Module Interrupt	V _{ref}	Optional		1	Conn1
CONN1_MODPRS#	Input	QSFP-DD Connector 1 Module Present	V _{ref}	Optional		1	Conn1
CONN1_MODSEL#	Output	QSFP-DD Connector 1 Module Select	V _{ref}	Optional		1	Conn1
CONN1_RESET#	Output	QSFP-DD Connector 1 Module Reset	V _{ref}	Optional		1	Conn1
CONN1_GREEN_LED	Output	QSFP-DD Connector 1 GREEN STATUS LED	V _{ref}	Optional		1	Conn1
CONN1_YELLOW_LED	Output	QSFP-DD Connector 1 YELLOW STATUS LED	V _{ref}	Optional		1	Conn1
CONN2_I_NITMODE	Output	QSFP-DD Connector 2 Module Initialization mode	V _{ref}	Optional		1	Conn1
CONN2_I_NT#	Input	QSFP-DD Connector 2 Module Interrupt	V _{ref}	Optional		1	Conn1
CONN2_MODPRS#	Input	QSFP-DD Connector 2 Module Present	V _{ref}	Optional		1	Conn1
CONN2_MODSEL#	Output	QSFP-DD Connector 2 Module Select	V _{ref}	Optional		1	Conn1
CONN2_RESET#	Output	QSFP-DD Connector 2 Module Reset	V _{ref}	Optional		1	Conn1
CONN2_GREEN_LED	Output	QSFP-DD Connector 2 GREEN STATUS LED	V _{ref}	Optional		1	Conn1
CONN2_YELLOW_LED	Output	QSFP-DD Connector 2 YELLOW STATUS LED	V _{ref}	Optional		1	Conn1
PRSNT0#	Output	Module present pin connector 0. Tied to GND through 1K resistor on module side.	GND	Required		1	Conn0

PRSNT1#	Output	Module present pin connector 1. Tied to GND through 1K resistor on module side	GND	Required		1	Conn1
SCALE_DEBUG_EN	Output	At-a-scale debug enable on the module. Isolates any motherboard JTAG debug path when logic high	3.3V	Required		1	Conn1
DEBUG_PORT_PRSENT#	Input	Presence signal for debug port in motherboard. Notifies logic in the module the debug access is being used by the motherboard debug connector. Debug port on baseboard present when logic low	GND	Required		1	Conn1
MNGMT_LINK0Tp/n	Output	Vendor specific module to module management link port 0 transmit. Required for some accelerators. Check with accelerator vendor.		Required	1		Conn1
MNGMT_LINK0Rp/n	Input	Vendor specific module to module management link port 0 receive. Required for some accelerators. Check with accelerator vendor.		Required	1		Conn1
MNGMT_LINK1Tp/n	Output	Vendor specific module to module management link port 1 transmit. Required for some accelerators. Check with accelerator vendor.		Required	1		Conn1
MNGMT_LINK1Rp/n	Input	Vendor specific module to module management link port 1 receive. Required for some accelerators.		Required	1		Conn1

		Check with accelerator vendor.					
TEST0	Input		V_{ref}	Optional		1	Conn0
TEST[4:1]	Input		V_{ref}	Optional		4	Conn0
TEST5	I/O		V_{ref}	Optional		1	Conn0
TEST6	I/O		V_{ref}	Optional		1	Conn0
TEST7	Output		V_{ref}	Optional		1	Conn0
TEST8	Input		V_{ref}	Optional		1	Conn0
TEST9	Input		V_{ref}	Optional		1	Conn0
TEST10	Input		V_{ref}	Optional		1	Conn1
TEST11	Output		V_{ref}	Optional		1	Conn1
TEST12	Input		V_{ref}	Optional		1	Conn1
TEST13	Input		V_{ref}	Optional		1	Conn1
TEST14	Input		V_{ref}	Optional		1	Conn1
MANF_MODE#	Input	Manufacturing Mode 1: Normal operation 0: Module enter into manufacturing mode	3.3V	Optional		1	Conn0
FW_RECOVERY#	Input	On board manageability boot recovery mode 1: Normal operation 0: Firmware Recovery boot mode	3.3V	Required		1	Conn0
TEST_MODE#	Input	Compliance Test Mode 1: Normal operation 0: ASIC/GPU enter into electrical compliance mode	V_{ref}	Required		1	Conn0
RFU[3:0]		Reserved for future use				4	Conn0
RFU[24:4]		Reserved for future use				21	Conn1

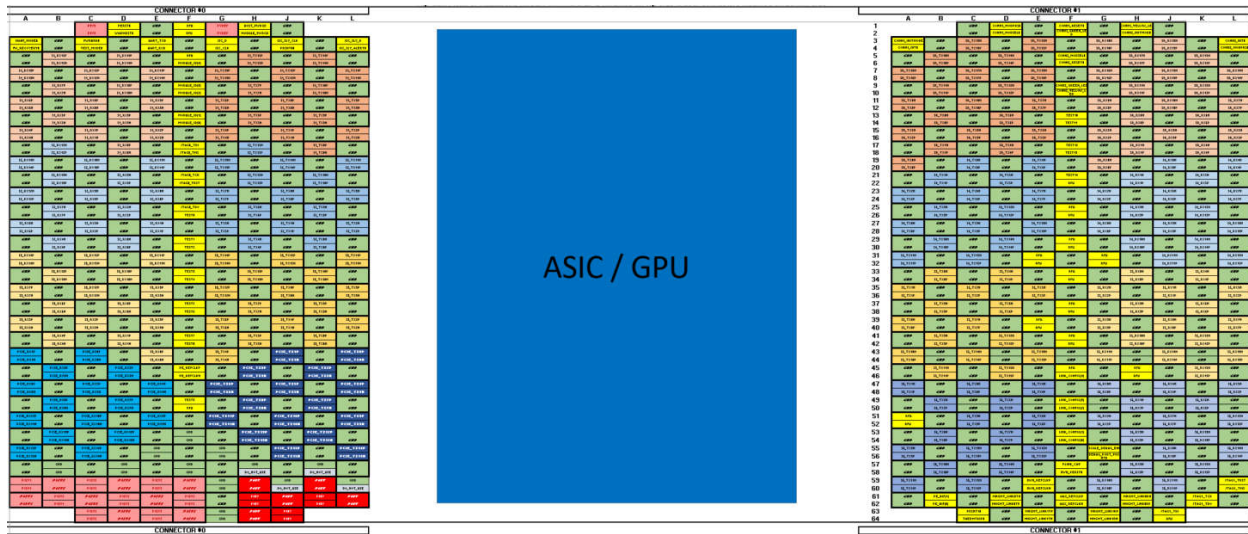


Figure 33 – Mezzanine Connector Pin map

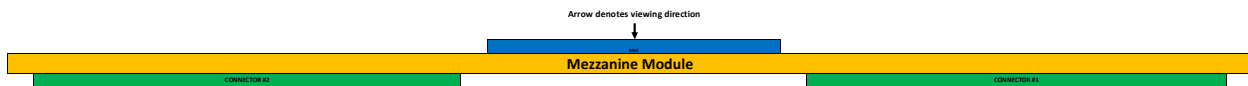


Figure 34 – Mezzanine Connector Pin Out View Reference

8.4 Accelerator Module Power Profiles

This section defines the maximum thermal design power (TDP) the module can support as well the excursion design power (EDP).

8.4.1 Thermal Design Power TDP

The module supports up to 350W TDP if the input nominal voltage is 12V. The module supports up to 700W if the input nominal voltage is 48V or 54V.

The accelerator baseboard supplies power to the module through the Mirror Mezz Connector0 power pins. There are 3 power rails defined in this document to accommodate both 12V and 48V (or 54V) modules. The current capability and power status are as the table below. The power is available on state S0 only. Only five P12V power pins are mandatory when the supply power is 48V (16 pins), and the rest of the P12V pins can be NC. When the baseboard supply power is 12V, P48V can be NC. The baseboard can supply all 3 power rails and supports both 12V and 48V modules.

Table 5 Power Rails

Power Rail	Voltage Tolerance	# of pins	Current Capability	Status
P12V	11V min to 13.2V max	27	27A (when at 11V)	Normal Power
P12V Mandatory	11V min to 13.2V max	5	5A (when at 11V)	Normal Power
P48V	44V min to 60V max	16	16A (when at 44V)	Normal power
P3.3V	3.3V ± 10%(max)	2	2A	Normal power

Note: To support even higher TDP accelerator modules, we can further bypass 12V to provide more 48V and vice versa. Baseboard designers should consider module detection when mixing different modules in the same system.

8.4.2 Excursion Design Power EDP

System baseboard designers should be sure to support the accelerator module’s excursion design power (aka EDP). The accelerator module VR electrical design must be designed to handle the instantaneous peak power short period (usually it is on the order of a μ s) with low duty cycle. The VR’s thermal design should be robust enough to handle lower power EDP level (e.g. 1.1x TDP) for ms level interval without asserting VR HOT over temperature alert. The system integrator should work with the module suppliers closely to ensure that the system baseboard supplies enough power to the module without triggering under voltage protection.

Table 6 Excursion Design Power Example

EDP	Duration
2x TDP	$\leq 20\mu$ s
1.6x TDP	≤ 2 ms
1.5x TDP	≤ 5 ms
1.2x TDP	≤ 10 ms
1.1x TDP	≤ 20 ms

8.5 System power sequencing

System designers should follow the below power sequence requirement to implement the design. It is recommended to check with each specific module specification to ensure the modules work properly.

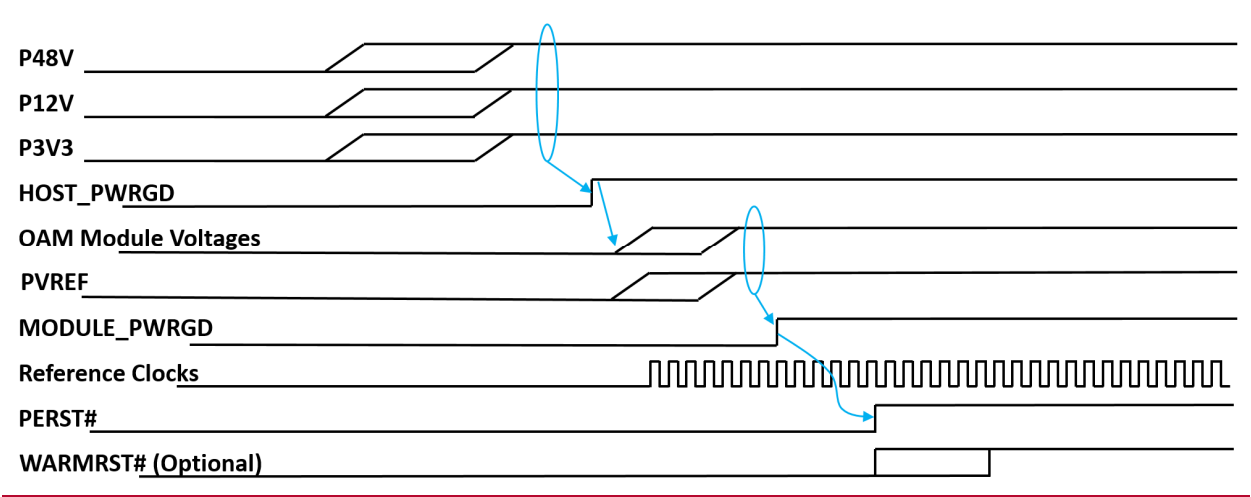


Figure 35 OCP Accelerator Module Power Sequence

Notes:

- 1) If the accelerator modules with the baseboard are in the disaggregated design from the host system, the HOST_PWRGD is the baseboard power good indication signal.

- 2) All voltages on the baseboard that OAM plugs into must be within specification before HOST_PWRGD is asserted;
- 3) HOST_PWRGD is the enable signal to the voltage regulators on the OAM;
- 4) As the voltage planes on the module ramp up, the reference clocks from the baseboard will begin to run;
- 5) After all the voltages on the module are within specification, the module asserts MODULE_PWRGD to the baseboard;
- 6) At least 100ms after Module_PWRGD assertion, the baseboard will de-assert the PCIe reset signal(PERST#) to the module.
- 7) The optional WARMRST# signal de-asserts at the same time or later than the PERST# signal is de-asserted.

8.6 Mezzanine Module Insertion Loss

The module interconnection channel total insertion loss at 14.025Ghz should not be over -8dB. The system integrator may contact the module supplier for details about the interconnection channel insertion loss and plan system baseboard design accordingly. For other speed is over 14.025Ghz, the system integrator should work with module supplier closely to determine the loss budget on the module as well as on the baseboard.

Total Tx or Rx loss on Module +mated Mezz Connectors @14.025Ghz	up to 8dB
---	-----------

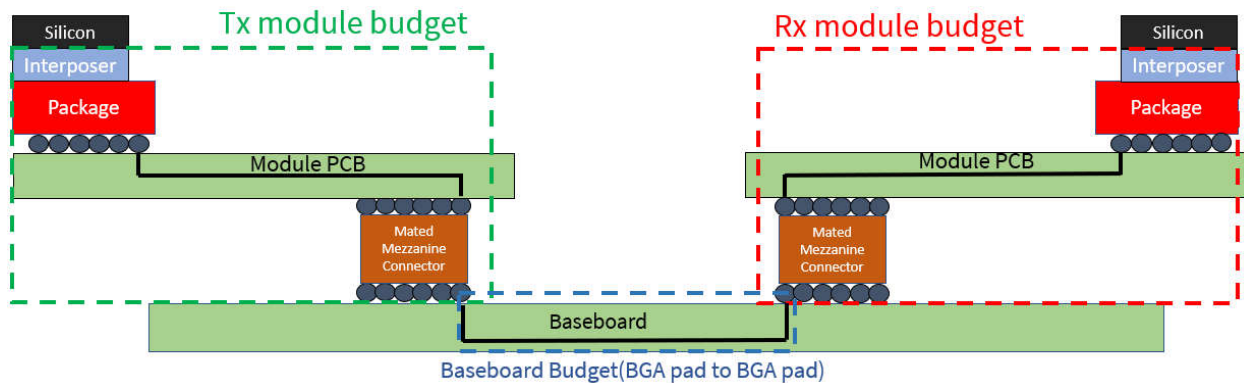


Figure 36 Channel loss Budget (PCB-only Topology)

Total channel loss budget = Tx module + baseboard + Rx module

Example:

Total channel loss budget = 30 dB (14.025 GHz)

- Tx module = 8 dB
- Rx module = 8 dB
- Baseboard budget = 30 dB – 8 dB – 8 dB = 14 dB

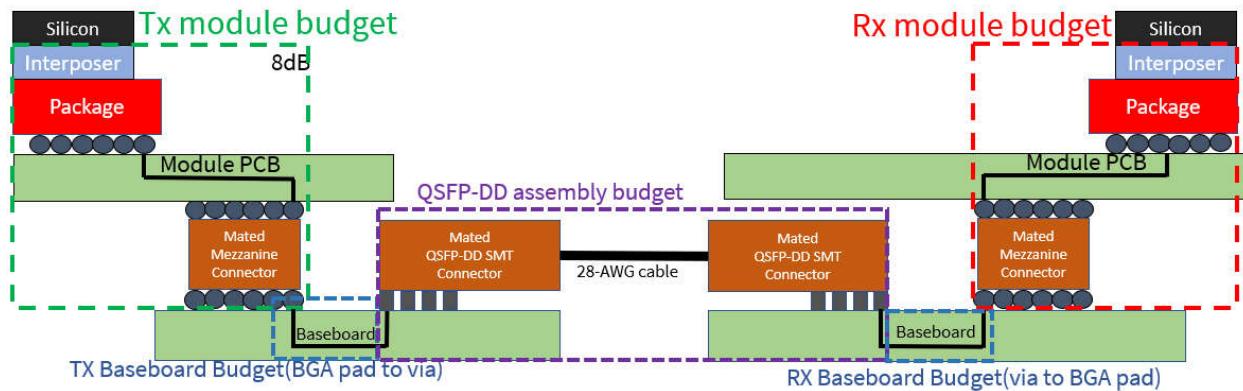


Figure 37 Channel loss budget (Cabled Topology)

Total channel loss budget = Tx module + Tx baseboard + QSFP-DD assembly + Rx baseboard + Rx module

Example:

Total channel loss budget = 30 dB (14.025 GHz)

- Tx module = 8 dB
- Rx module = 8 dB
- QSFP-DD = 5 dB
- Tx + Rx baseboard budget = 30 dB – 8 dB – 8 dB – 5.0 dB = 9 dB

8.7 Management link

Management link 0/1 are defined for OAM to communicate with each other. When host interface is PCIe, management links are routed in a ring as illustrated in the following diagrams with Management link 0 on one module connecting to management link 1 on the next module.

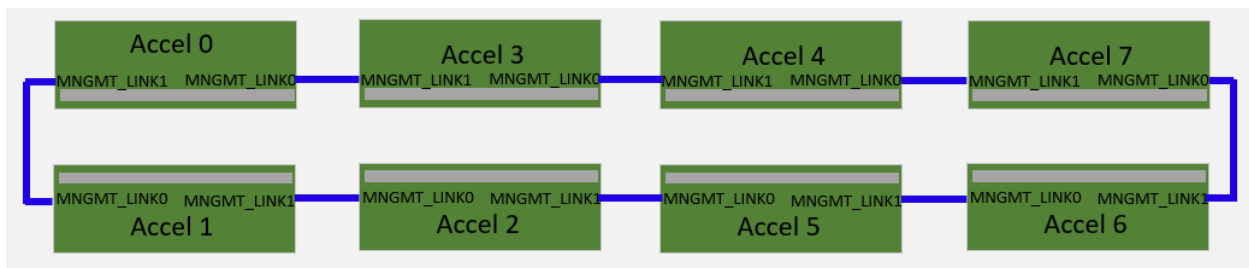


Figure 38 Management Link Routing Guidance

9 Mezzanine Module Interconnect Topologies

This section describes the recommended interconnection topology for a system with 4, 8, and 16 accelerators.

9.1 Module ID

The following figure shows the MODULE_ID[4:0] strapping for physical orientation of modules when 8 interconnected Accelerators are used.

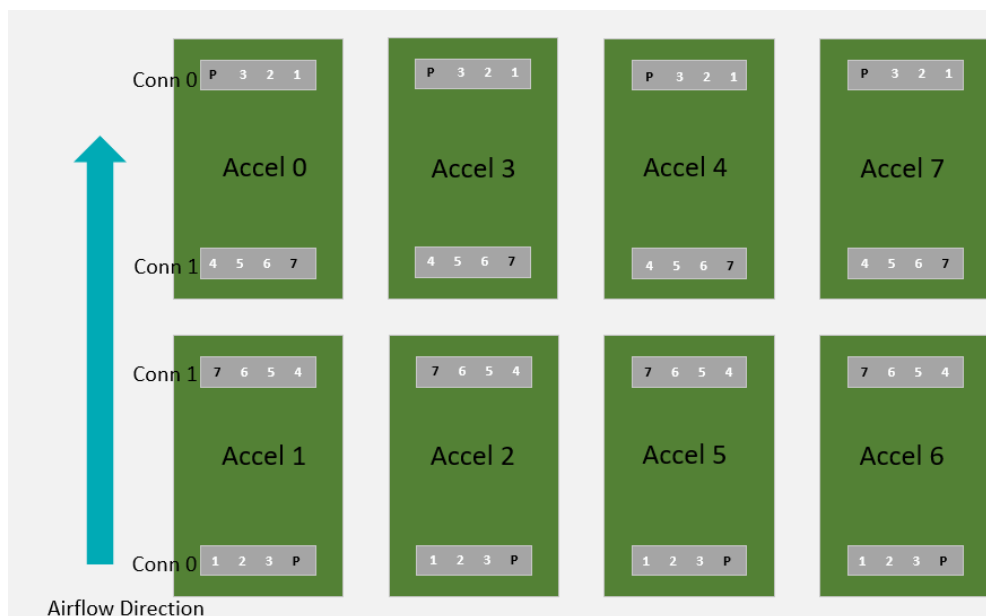


Figure 39 Required MODULE_ID[4:0] assignments for motherboards with 8 interconnected modules

Detail port to port assignment is based on system placement and routing length. Module to module interconnect may decrease to 4 ports if the module only supports 4. Module to module interconnect link may only utilize 8 lanes if the module defines 8 lanes per link.

MODULE_ID can be used as the I2C address strap pins if needed.

The following Figure shows the required MODULE_ID[4:0] assignments when only 4 modules are connected as two rows of two.

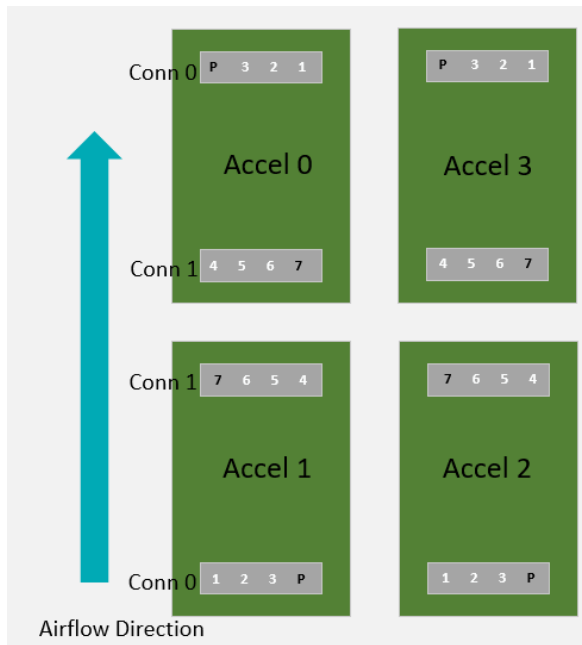


Figure 40 MODULE_ID[4:0] assignments when four only in two rows of two

9.2 Interconnect Topology

This section describes different interconnect topologies and routing guidance for 8 OAMs with different port numbers. If all 7 ports are configured and routed as x16, there is no additional port(s) for expansion. To reserve expansion port(s), we suggest limiting on board Interconnect links up to x8. 2nd half of port 1(x8, also referred as 1H) is reserved for expansion by default. 2nd half of port 4,6,7 are used in X8 based full connected topology at section 9.2.4.

9.2.1 Hybrid Cube Mesh (HCM) for 6 ports per OAM

Below Figure shows an example topology (Hybrid Cube Mesh) of 8 modules in a baseboard.

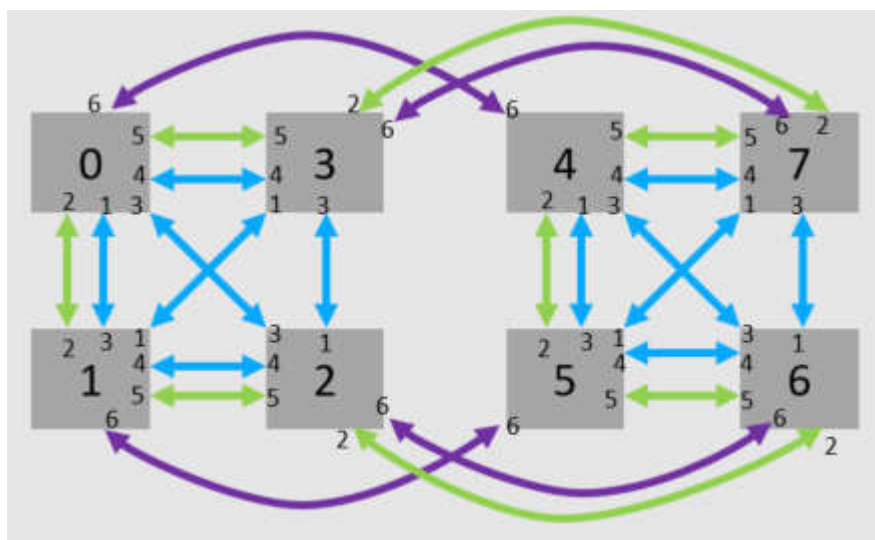


Figure 41 Topology Example for Modules with 3/4/6 ports – Hybrid Cube Mesh

The interconnect topology in this Figure supports the following OAM module interconnects:

- 3 links and two fully connected quads using links: 1, 3, and 4
- 4 link Hybrid Cube Mesh using links: 1, 3, 4, and 6
- 6 link Hybrid Cube Mesh using links: 1, 2, 3, 4, 5, and 6

Here is the routing suggestion for Hybrid Cube Mesh:

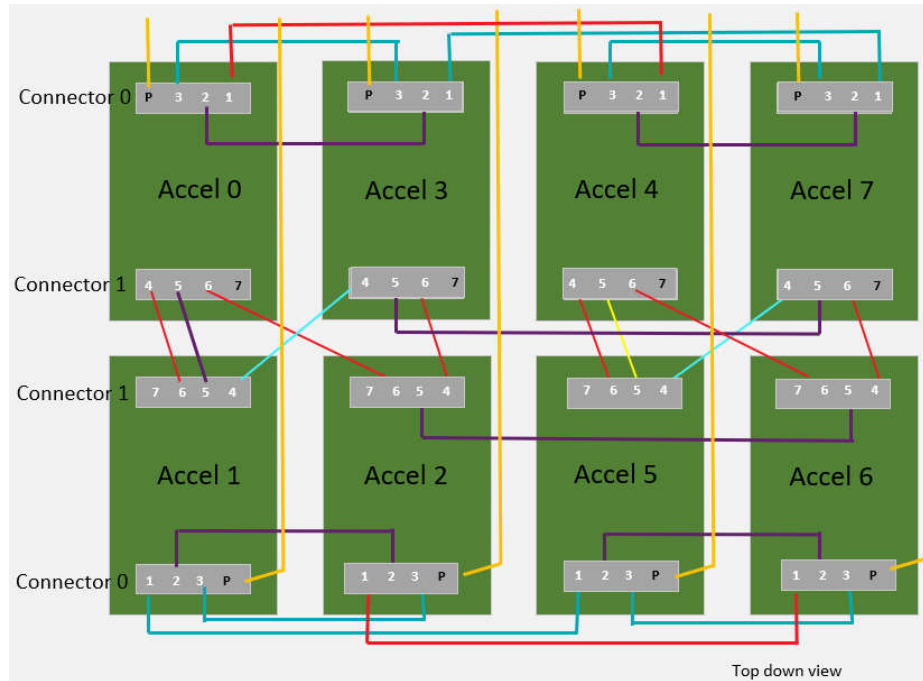


Figure 42 6 port Hybrid Cube Mesh Routing Recommendation

9.2.2 Almost Fully Connected

Depending on different workloads, if the module has 6 links, the other topology that can be considered is Almost Fully Connected (also called as Chordal Ring). Each module connects to the other 6 of the modules with 1 link. Figure below illustrates the topology:

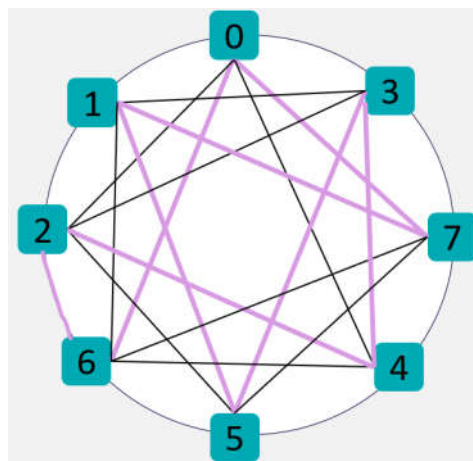


Figure 43 Almost Fully Connected Topology

The interconnect topology in below Figure supports the following OAM module interconnects:

- 6 x16 links Chordal Ring (Almost Fully Connected) using links: 1, 2, 3, 4, 5, and 6

Here is the routing suggestion:

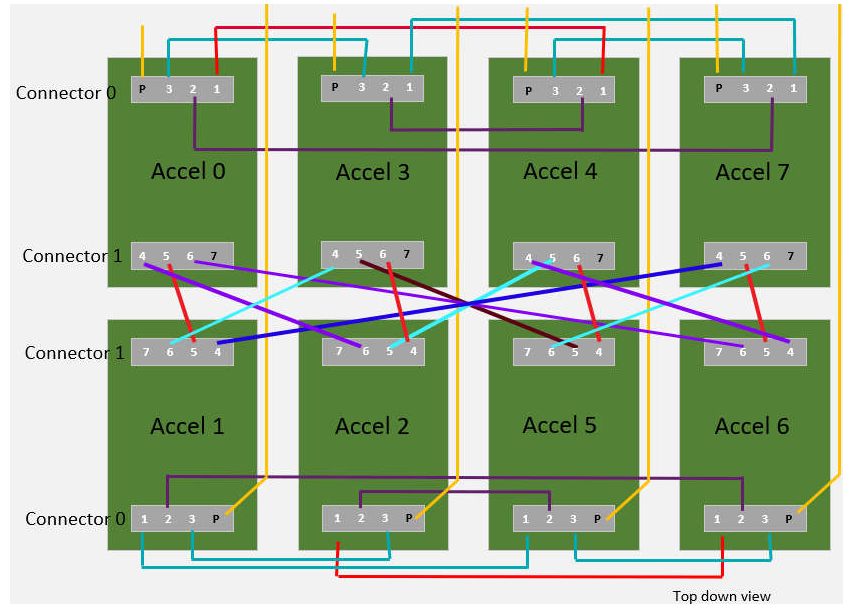


Figure 44 Routing Recommendation for Almost Fully Connected Topology

9.2.3 Hybrid Cube Mesh for 8 ports per OAM

The Figure below shows an example of 8 ports topology (Hybrid Cube Mesh) of 8 modules in a baseboard. Please follow port mapping to design OAM in order to be able to fit in the universal OAM baseboard.

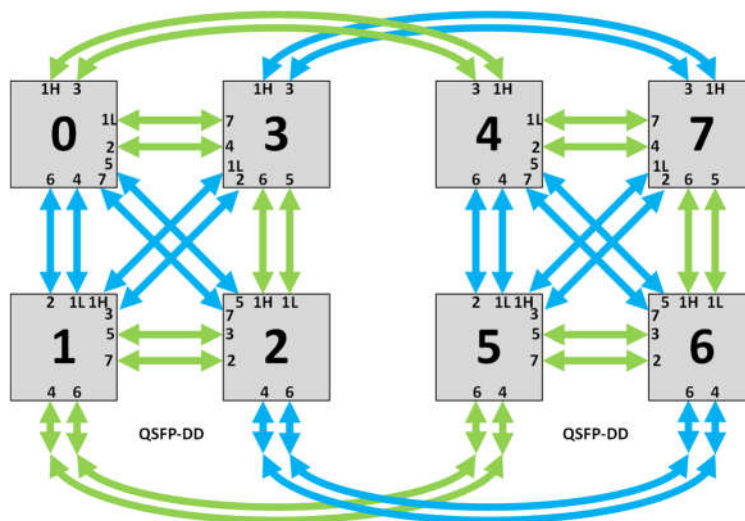


Figure 45 8-port Hybrid Cube Mesh Topology

And here is routing suggestion: total 4 layer, two layers for TX two layers for RX. Port 4/6 are connected through cables.

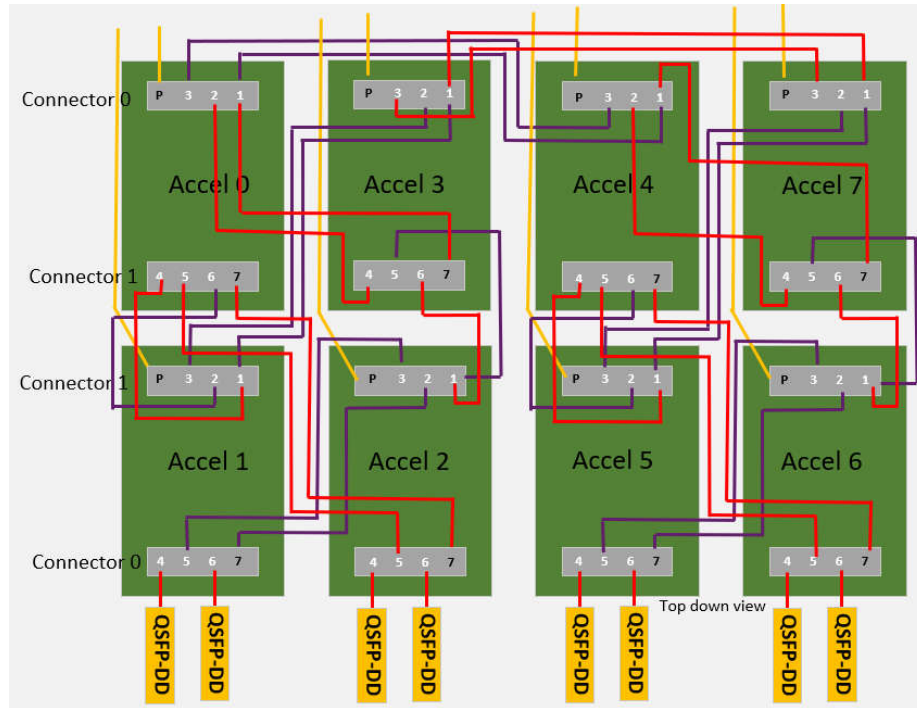


Figure 46 8-port HCM topology routing guide

9.2.4 Fully Connected

If the module has 7 or more links, each module can communicate with any of the other 7 modules directly. The topology is fully connected.

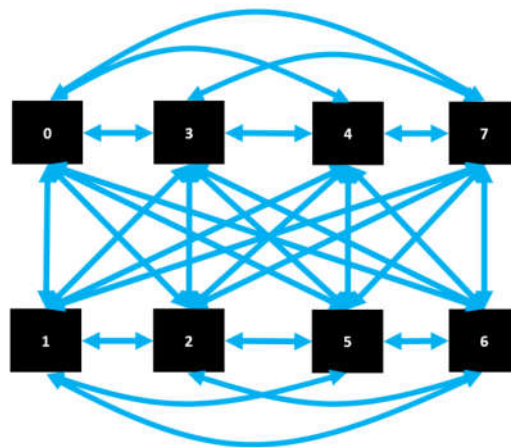


Figure 47 Fully Connected Topology

For fully connect with expansion consideration, the baseboard link is suggested to route as X8 with some extra ports(2nd X8s) for expansion for embedding other topologies. Here is 7X8 links fully connected routing example:

- Fully connected: 7 x8 links using port 1-7 first X8(1L-7L);
- 2nd half of port 1(1H) can be used for expansion
- HCM: X16 link for port 4/6, X8 link(5L) for port 5 and 2nd half of port 7(7H)

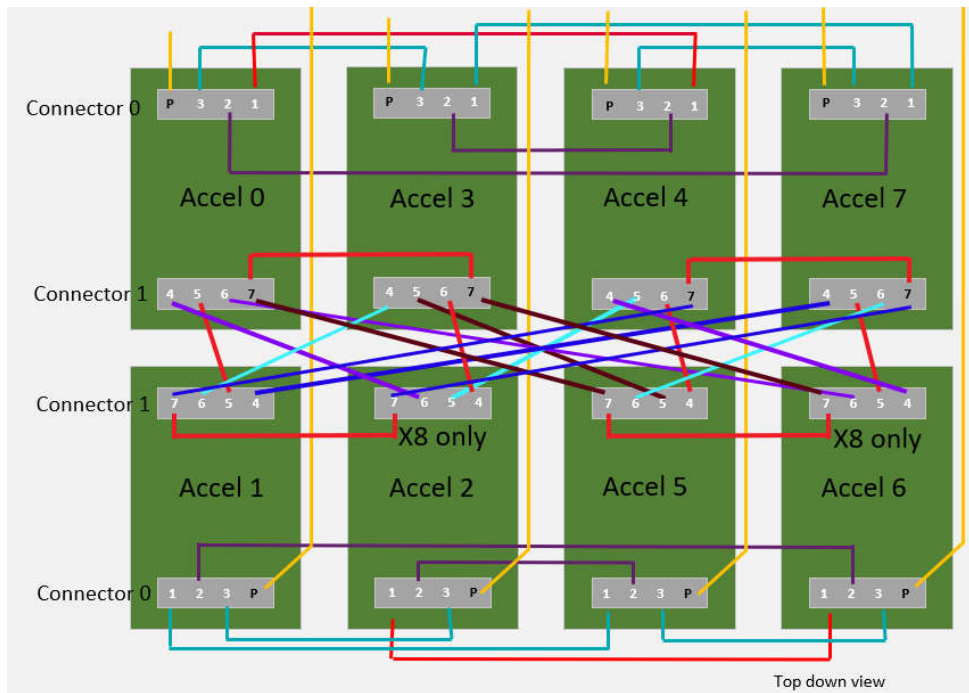


Figure 48 Routing Recommendation for Fully Connected Topology

9.2.5 4D Hypercube

16 modules with 6 links per module the interconnect topology could be a 4d hypercube:

- Four fully connected quads
- Each quad connected to the other 3 quads at all four corners
- The green links below matches the green line in the 4d hypercube

As one single PCB cannot fit all 16 modules, this topology interconnect will have cable or backplane to connect between PCBs. Depending on whether one single PCB holds 4 modules or 8 modules, the interconnect path may be different. The system integrator may discuss with the module supplier for details. The module defines two QSFPDD ports as the potential scale up solution.

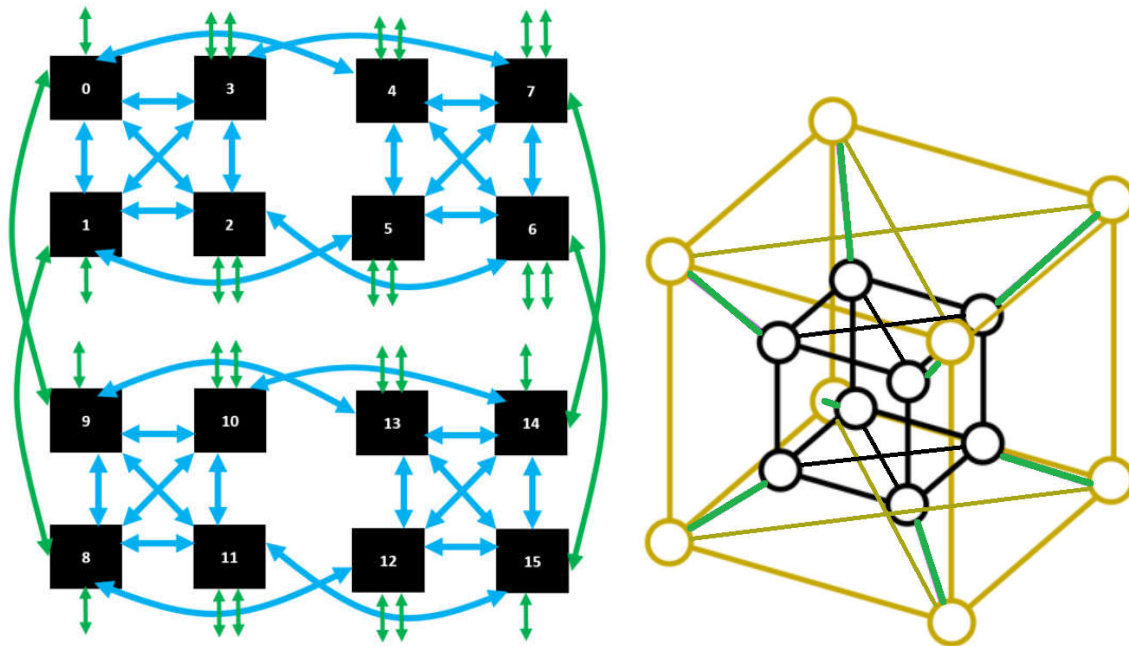


Figure 49 4D Hypercube

9.3 LINK_CONFIG[4:0]

The 5 link configuration strapping bits are pulled up on modules that use them. These bits are strapped to ground on the baseboard to select logic 0, or left floating on the baseboard to select logic 1. Some accelerators use these LINK_CONFIG[4:0] strapping bits to determine the interconnect topology for the links between modules and to determine the protocol of the “P” Link and the protocol and function of the optional “R” Link.

Encodings not listed in the table below are currently un-defined.

Table 7 LINK_CONFIG[4:0] Encoding Definitions

LINK_CONFIG[4:0]	Definition
00000	Reserved for Accelerator Test use by Accelerator Vendor.
xxxx0 (except for 00000)	Indicates the “P” link is PCIe
01000	6 link HMC, 4 link HMC, and two 3 link fully connected quads as connected in Figure 41.
00100	6 x16 almost fully connected
01010	4 or 6 x8 link HMC, 7x8 links fully connected, and two 3 x8 link fully connected quads as connected in Figure 48.
01100	8 link HMC as in Figure 45.
xxxx1 (except for 11111)	Indicates the “P” link is an alternate protocol other than PCIe.

11111	Indicates an alternate means for identifying the link interconnect topology and configuration is used.
-------	--

9.4 Accelerator Interconnect PCB Topology

It is recommended to keep the stub for interconnect link less than 10mils. System integrator should work with module suppliers to plan the PCB routing and address the signal integrity concern carefully.

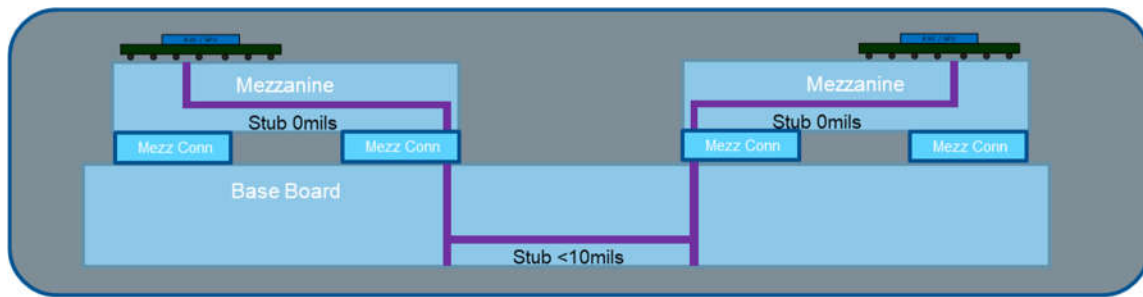


Figure 50 – Accelerator Interconnect PCB Topology

10 Mezzanine module reference system design

This section gives a system design concept as a reference. Figure 51 Reference System Design shows 8 OAM modules. The plastic top provides a 0.5mm bumper on each side of the 102mm width OAM, and the 1mm gap between each module assembly provides rough alignment, guidance, and keying as described in Section 6.6. An air baffle is designed into the 33.8mm space to prevent air bypass in the system. Note that the front and rear rows are oriented 180 degrees opposite as indicated in Section 9.Recommended Alignment Features

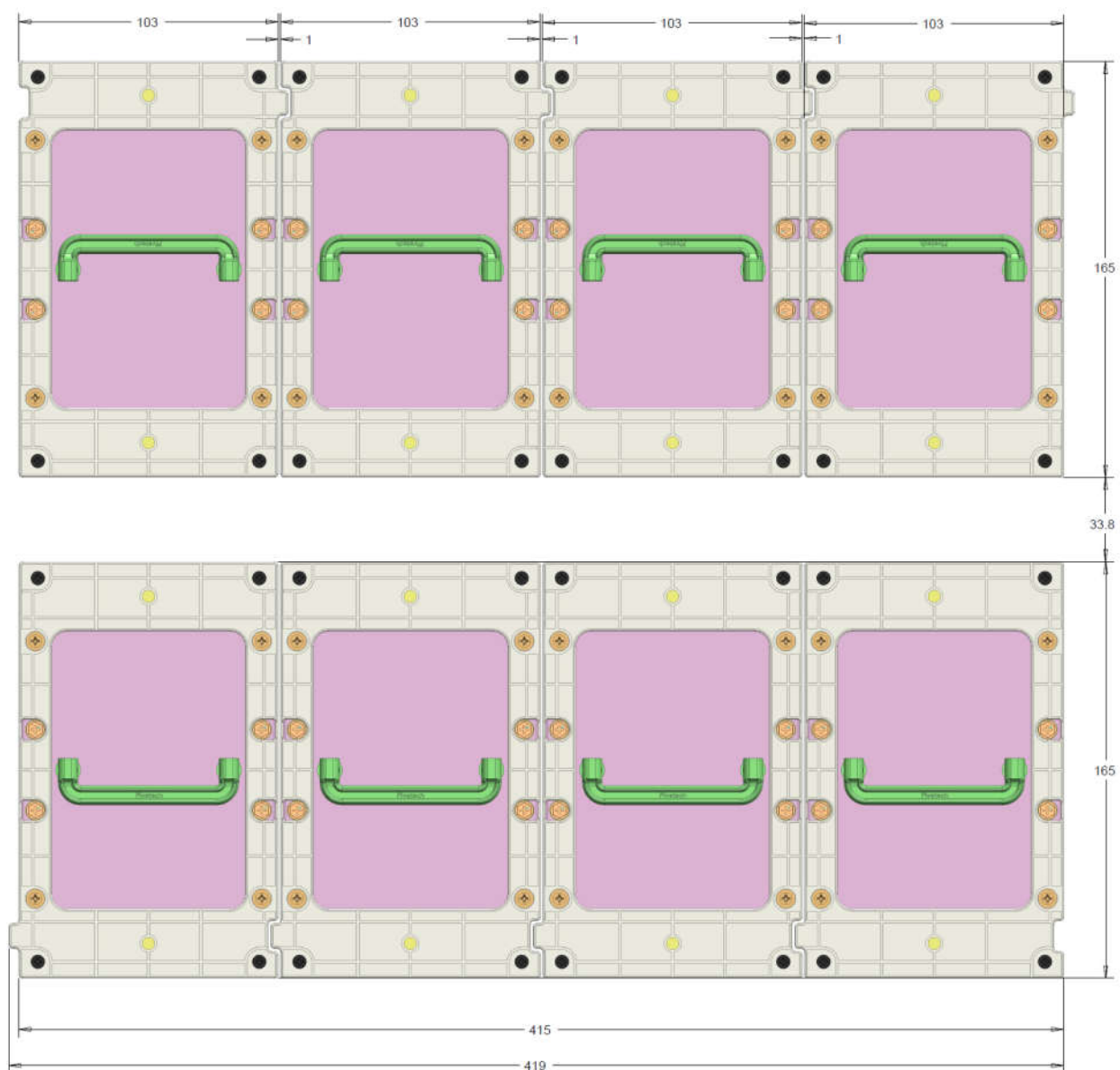


Figure 51 Reference System Design

11 System Level Communication

This section describes a common set of management requirements for this OAM.

11.1 Sideband Management Interface

The OCP Accelerator Mezz module sideband management interface is used by a Management Controller (MC) or Baseboard Management Controller (BMC) to communicate with the Accelerator module. Table 8 below summarizes the sideband management interfaces.

Table 8 Sideband Management Interface

Requirement	Voltage Level	Required
SMBus 2.0/I2C compliant physical interface(primary)	3.3V	Yes
JTAG physical interface(primary)	LV I/O	Yes
SMBus 2.0/I2C compliant physical interface(secondary)	LV I/O	Optional
JTAG physical interface(secondary)	3.3V	Optional

The OAM communicates with the Baseboard Management Controller (BMC) via I2C/SMBus by using:

- Standard Intelligence Platform Management Bus (IPMB) and Intelligent Platform Management Interface (IPMI) commands.
- I2C command

11.2 Sensor Reporting

An OAM module may have several silicon components including one or more ASICs implementing acceleration functions. For the system management, it is important that related sensors (voltage, current/power, temperature...etc) of these components can be retrieved over sideband interfaces.

The sensor reporting interface will only be accessible in main power mode (S0). Table 9 summarizes the sensors reporting list. The report from these sensors improves the system monitoring/management and allows the baseboard management device to access key components on the module. It is recommended that the voltage/current/power sensor reporting accuracy is within $\pm 1\%$ and the temperature reporting accuracy is within $\pm 3^{\circ}\text{C}$.

Table 9 Sensor List

Sensor List	Remark
Power/Current (Mem, core, module level)	
Voltage (core,mem)	
ASIC Temp (hot spot and edge temp)	
ASIC T_{jmax}	
Mem Temp for each stack	
Core Power VR Temp	
Core Power VR Vol	

Inlet sensor	Need define common location
Outlet sensor	

11.3 Error Monitoring/Reporting

System Management Controller (MC) or Baseboard Management Controller (BMC) shall be able to monitor and access the accelerator module's internal registers as needed to set thresholds, clear status, determine error counts and syndromes (SW driven interrupt or an Alert pin), and identify error sources/Syndromes etc.

- Mem
- PCIe
- ASIC
- Interconnect Links
- Others, I/O

11.4 Firmware Update

The accelerator module should support secure boot. The ASIC(s) needs to check signature of the firmware each time it boots against keys programmed in OTP.

- It will check using public keys burned into the silicon.
- If the signature does not match, the chip will not boot.
- Enter some sort of recovery mode allowing firmware to be flashed by the host.
- Time-of-check-time-of-use (TOCTOU) attacks must be prevented. Specifically, host should NOT be able to modify firmware image once signature checks begin, and should also be unable to modify the stored or running firmware image after signature checks have been completed.
- Key-revocation and anti-rollback protection must be provided for the accelerator firmware.
- Protection against PDoS attacks must be provided.

Module supplier should provide the relevant tool for out-of-band firmware updates.

11.5 Power Capping

Module supplier should provide either the tool or I2C/IPMI commands to do power capping for the module.

11.6 FRU Information

System Management Controller (MC) or Baseboard Management Controller (BMC) shall be able to access related internal registers to get module information, such as vendor ID, device ID, Manufacturing Serial number, firmware version, Bifurcation information, etc.

11.7 IO Calibration

System Management Controller (MC) or Baseboard Management Controller (BMC) shall be able to get DDR/PCIe/interconnect training status and margin information. Special tools and API may need to be provided by module vendors.

12 Environmental

12.1 Environmental Requirements

The OAM shall meet the following environmental requirements:

- Gaseous Contamination: Severity Level G1 per ANSI/ISA 71.04-1985
- Ambient operating temperature range: 5°C to +35°C
- Operating and Storage relative humidity: 20% to 90% (non-condensing)
- Storage temperature range: -20°C to +70°C
- Transportation temperature range: -55°C to +85°C (short-term storage)
- Operating altitude with no de-ratings: 3048m (10000 feet) – recommended as this is a Facebook spec and standard for Telco operation

12.2 Regulation

The vendor needs to provide CB reports of the mezzanine module. These documents are needed to have rack level CE. The OAM should be compliant with RoHS and WEEE. The PCB should have a UL 94V-0 certificate.

13 Revision History

Author	Description	Revision	Date
Whitney Zhao	Initial Release	0.1	6/29/2018
Whitney Zhao	Add pin list	0.2	11/09/2018
Tiffany Jin Cheng Chen	Update Mechanical, Thermal, Reliability sections	Internal release	1/24/2019
Whitney Zhao Siamak Tavallaei	Add power profiles, power sequence requirement Update license information, overview	Internal release	2/4/2019
Tiffany Jin Whitney Zhao	Update module drawings Update pin list, recommended topologies	0.8	2/7/2019
Whitney Zhao	Updated some typos, interconnect topologies Add LINK_CONFIG table	0.82	
Tiffany Jin	Update ME section	0.83	
Cheng Chen	Add liquid cooling concept drawing	0.84	
Whitney Zhao	Update license information. Update topology port mapping Add 156.25Mhz clock for Serdes	0.85	
Whitney Zhao Tiffany Jin Cheng Chen	Update AFC interconnect topology routing recommendation to be compatible with HMC and FC Add management link routing guidance Update SerDes pin map Update ME drawings; more detail on requirements vs recommendations Add reference data for the increasing need of airflow for OAM cooling as die power increases	0.90	
Whitney Zhao	Change SerDes R to 7, update it from X20 to X16 Add two power reduction GPIO pins PWRRDT#[1:0] Add 8-port HCM topology and routing guide Update Vref range to 1.5V-3.3V Update section 8.6, add channel loss budget detail diagrams Update OAM_Pin_map to rev1.0: Detail change list pls refer to OAM_Pin_map_rev1.0 spreadsheet; Update OAM_Pin_list to R1.0: detail change list pls refer to OAM_Pin_list_Rev1.0 spreadsheet	1.0	